

CHAPTER 11

Limits, Continuity and Differentiability

Section-A

JEE Advanced/ IIT-JEE

A Fill in the Blanks

1. Let $f(x) = \begin{cases} (x-1)^2 \sin \frac{1}{(x-1)} |x| & \text{if } x \neq 1 \\ -1, & \text{if } x = 1 \end{cases}$
be a real-valued function. Then the set of points where $f(x)$ is not differentiable is (1981 - 2 Marks)

2. Let $f(x) = \begin{cases} \frac{(x^3 + x^2 - 16x + 20)}{(x-2)^2}, & \text{if } x \neq 2 \\ k, & \text{if } x = 2 \end{cases}$
If $f(x)$ is continuous for all x , then $k = \dots\dots\dots$ (1981 - 2 Marks)

3. A discontinuous function $y = f(x)$ satisfying $x^2 + y^2 = 4$ is given by $f(x) = \dots\dots\dots$ (1982 - 2 Marks)

4. $\lim_{x \rightarrow 1} (1-x) \tan \frac{\pi x}{2} = \dots\dots\dots$ (1984 - 2 Marks)

5. If $f(x) = \sin x$, $x \neq n\pi$, $n = 0, \pm 1, \pm 2, \pm 3, \dots\dots\dots$
 $= 2$, otherwise
and $g(x) = x^2 + 1$, $x \neq 0, 2$
 $= 4, x = 0$
 $= 5, x = 2$,

then $\lim_{x \rightarrow 0} g[f(x)]$ is (1986 - 2 Marks)

6. $\lim_{x \rightarrow -\infty} \left[\frac{x^4 \sin\left(\frac{1}{x}\right) + x^2}{(1 + |x|^3)} \right] = \dots\dots\dots$ (1987 - 2 Marks)

7. If $f(9) = 9$, $f'(9) = 4$, then $\lim_{x \rightarrow 9} \frac{\sqrt{f(x)} - 3}{\sqrt{x} - 3}$ equals (1988 - 2 Marks)

8. ABC is an isosceles triangle inscribed in a circle of radius r . If $AB = AC$ and h is the altitude from A to BC then the triangle ABC has perimeter $P = 2(\sqrt{2hr} - h^2) + \sqrt{2hr}$ and area $A = \dots\dots\dots$ also $\lim_{h \rightarrow 0} \frac{A}{P^3} = \dots\dots\dots$ (1989 - 2 Marks)

9. $\lim_{x \rightarrow \infty} \left(\frac{x+6}{x+1} \right)^{x+4} = \dots\dots\dots$ (1990 - 2 Marks)

10. Let $f(x) = x |x|$. The set of points where $f(x)$ is twice differentiable is (1992 - 2 Marks)

11. Let $f(x) = [x] \sin \left(\frac{\pi}{[x+1]} \right)$, where $[\bullet]$ denotes the greatest integer function. The domain of f is... and the points of discontinuity of f in the domain are..... (1996 - 2 Marks)

12. $\lim_{x \rightarrow 0} \left(\frac{1+5x^2}{1+3x^2} \right)^{1/x^2} = \dots\dots\dots$ (1996 - 1 Mark)

13. Let $f(x)$ be a continuous function defined for $1 \leq x \leq 3$. If $f(x)$ takes rational values for all x and $f(2) = 10$, then $f(1.5) = \dots\dots\dots$ (1997 - 2 Marks)

B True / False

1. If $\lim_{x \rightarrow a} [f(x)g(x)]$ exists then both $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exist. (1981 - 2 Marks)

C MCQs with One Correct Answer

1. If $f(x) = \sqrt{\frac{x - \sin x}{x + \cos^2 x}}$, then $\lim_{x \rightarrow \infty} f(x)$ is (1979)
(a) 0 (b) ∞
(c) 1 (d) none of these

2. For a real number y , let $[y]$ denotes the greatest integer less than or equal to y : Then the function $f(x) = \frac{\tan(\pi[x - \pi])}{1 + [x]^2}$ is (1981 - 2 Marks)
(a) discontinuous at some x
(b) continuous at all x , but the derivative $f'(x)$ does not exist for some x
(c) $f'(x)$ exists for all x , but the second derivative $f''(x)$ does not exist for some x
(d) $f'(x)$ exists for all x

3. There exist a function $f(x)$, satisfying $f(0) = 1, f'(0) = -1, f(x) > 0$ for all x , and (1982 - 2 Marks)
 (a) $f''(x) > 0$ for all x (b) $-1 < f''(x) < 0$ for all x
 (c) $-2 \leq f''(x) \leq -1$ for all x (d) $f''(x) < -2$ for all x
4. If $G(x) = -\sqrt{25 - x^2}$ then $\lim_{x \rightarrow 1} \frac{G(x) - G(1)}{x - 1}$ has the value (1983 - 1 Mark)
 (a) $\frac{1}{24}$ (b) $\frac{1}{5}$
 (c) $-\sqrt{24}$ (d) none of these
5. If $f(a) = 2, f'(a) = 1, g(a) = -1, g'(a) = 2$, then the value of $\lim_{x \rightarrow a} \frac{g(x)f(a) - g(a)f(x)}{x - a}$ is (1983 - 1 Mark)
 (a) -5 (b) $\frac{1}{5}$
 (c) 5 (d) none of these
6. The function $f(x) = \frac{\ln(1+ax) - \ln(1-bx)}{x}$ is not defined at $x = 0$. The value which should be assigned to f at $x = 0$ so that it is continuous at $x = 0$, is (1983 - 1 Mark)
 (a) $a - b$ (b) $a + b$
 (c) $\ln a - \ln b$ (d) none of these
7. $\lim_{n \rightarrow \infty} \left\{ \frac{1}{1-n^2} + \frac{2}{1-n^2} + \dots + \frac{n}{1-n^2} \right\}$ is equal to (1984 - 2 Marks)
 (a) 0 (b) $-\frac{1}{2}$
 (c) $\frac{1}{2}$ (d) none of these
8. If $f(x) = \frac{\sin[x]}{[x]}$, $[x] \neq 0$ (1985 - 2 Marks)
 $= 0, [x] = 0$
 Where $[x]$ denotes the greatest integer less than or equal to x . then $\lim_{x \rightarrow 0} f(x)$ equals -
 (a) 1 (b) 0
 (c) -1 (d) none of these
9. Let $f: R \rightarrow R$ be a differentiable function and $f(1) = 4$. Then the value of $\lim_{x \rightarrow 1} \int_4^{f(x)} \frac{2t}{x-1} dt$ is (1990 - 2 Marks)
 (a) $8f'(1)$ (b) $4f'(1)$ (c) $2f'(1)$ (d) $f'(1)$
10. Let $[.]$ denote the greatest integer function and $f(x) = [\tan^2 x]$, then: (1993 - 1 Mark)
 (a) $\lim_{x \rightarrow 0} f(x)$ does not exist
 (b) $f(x)$ is continuous at $x = 0$
 (c) $f(x)$ is not differentiable at $x = 0$
 (d) $f'(0) = 1$
11. The function $f(x) = [x] \cos\left(\frac{2x-1}{2}\right)\pi$, $[.]$ denotes the greatest integer function, is discontinuous at (1995S)
 (a) All x (b) All integer points
 (c) No x (d) x which is not an integer
12. $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{2n} \frac{r}{\sqrt{n^2 + r^2}}$ equals (1997 - 2 Marks)
 (a) $1 + \sqrt{5}$ (b) $-1 + \sqrt{5}$ (c) $-1 + \sqrt{2}$ (d) $1 + \sqrt{2}$
13. The function $f(x) = [x]^2 - [x^2]$ (where $[y]$ is the greatest integer less than or equal to y), is discontinuous at (1999 - 2 Marks)
 (a) all integers
 (b) all integers except 0 and 1
 (c) all integers except 0
 (d) all integers except 1
14. The function $f(x) = (x^2 - 1) |x^2 - 3x + 2| + \cos(|x|)$ is NOT differentiable at (1999 - 2 Marks)
 (a) -1 (b) 0 (c) 1 (d) 2
15. $\lim_{x \rightarrow 0} \frac{x \tan 2x - 2x \tan x}{(1 - \cos 2x)^2}$ is (1999 - 2 Marks)
 (a) 2 (b) -2 (c) $1/2$ (d) $-1/2$
16. For $x \in R, \lim_{x \rightarrow \infty} \left(\frac{x-3}{x+2} \right)^x =$ (2000S)
 (a) e (b) e^{-1} (c) e^{-5} (d) e^5
17. $\lim_{x \rightarrow 0} \frac{\sin(\pi \cos^2 x)}{x^2}$ equals (2001S)
 (a) $-\pi$ (b) π (c) $\pi/2$ (d) 1
18. The left-hand derivative of $f(x) = [x] \sin(\pi x)$ at $x = k, k$ an integer, is (2001S)
 (a) $(-1)^k(k-1)\pi$ (b) $(-1)^{k-1}(k-1)\pi$
 (c) $(-1)^k k\pi$ (d) $(-1)^{k-1} k\pi$
19. Let $f: R \rightarrow R$ be a function defined by $f(x) = \max\{x, x^3\}$. The set of all points where $f(x)$ is NOT differentiable is (2001S)
 (a) $\{-1, 1\}$ (b) $\{-1, 0\}$ (c) $\{0, 1\}$ (d) $\{-1, 0, 1\}$
20. Which of the following functions is differentiable at $x = 0$? (2001S)
 (a) $\cos(|x|) + |x|$ (b) $\cos(|x|) - |x|$
 (c) $\sin(|x|) + |x|$ (d) $\sin(|x|) - |x|$
21. The domain of the derivative of the function

$$f(x) = \begin{cases} \tan^{-1} x & \text{if } |x| \leq 1 \\ \frac{1}{2}(|x| - 1) & \text{if } |x| > 1 \end{cases}$$
 is (2002S)
 (a) $R - \{0\}$ (b) $R - \{1\}$
 (c) $R - \{-1\}$ (d) $R - \{-1, 1\}$

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22. The integer n for which $\lim_{x \rightarrow 0} \frac{(\cos x - 1)(\cos x - e^x)}{x^n}$ is a finite non-zero number is (2002S)
 (a) 1 (b) 2 (c) 3 (d) 4
23. Let $f: R \rightarrow R$ be such that $f(1) = 3$ and $f'(1) = 6$. Then $\lim_{x \rightarrow 0} \left(\frac{f(1+x)}{f(1)} \right)^{1/x}$ equals (2002S)
 (a) 1 (b) $e^{1/2}$ (c) e^2 (d) e^3
24. If $\lim_{x \rightarrow 0} \frac{((a-n)x - \tan x) \sin nx}{x^2} = 0$, where n is nonzero real number, then a is equal to (2003S)
 (a) 0 (b) $\frac{n+1}{n}$ (c) n (d) $n + \frac{1}{n}$
25. $\lim_{h \rightarrow 0} \frac{f(2h+2+h^2) - f(2)}{f(h-h^2+1) - f(1)}$, given that $f'(2) = 6$ and $f'(1) = 4$
 (a) does not exist (b) is equal to $-3/2$
 (c) is equal to $3/2$ (d) is equal to 3 (2003S)
26. If $f(x)$ is differentiable and strictly increasing function, then the value of $\lim_{x \rightarrow 0} \frac{f(x^2) - f(x)}{f(x) - f(0)}$ is (2004S)
 (a) 1 (b) 0 (c) -1 (d) 2
27. The function given by $y = ||x| - 1|$ is differentiable for all real numbers except the points (2005S)
 (a) $\{0, 1, -1\}$ (b) ± 1 (c) 1 (d) -1
28. If $f(x)$ is continuous and differentiable function and $f(1/n) = 0 \forall n \geq 1$ and $n \in I$, then (2005S)
 (a) $f(x) = 0, x \in (0, 1]$
 (b) $f(0) = 0, f'(0) = 0$
 (c) $f(0) = 0 = f'(0), x \in (0, 1]$
 (d) $f(0) = 0$ and $f'(0)$ need not to be zero
29. The value of $\lim_{x \rightarrow 0} \left((\sin x)^{1/x} + (1+x)^{\sin x} \right)$, where $x > 0$ is (2006 - 3M, -1)
 (a) 0 (b) -1 (c) 1 (d) 2
30. Let $f(x)$ be differentiable on the interval $(0, \infty)$ such that $f(1) = 1$, and $\lim_{t \rightarrow x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$ for each $x > 0$. Then $f(x)$ is (2007 - 3 marks)
 (a) $\frac{1}{3x} + \frac{2x^2}{3}$ (b) $\frac{-1}{3x} + \frac{4x^2}{3}$ (c) $\frac{-1}{x} + \frac{2}{x^2}$ (d) $\frac{1}{x}$
31. $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\int_{\frac{\pi}{4}}^x \sec^2 t \, dt}{x^2 - \frac{\pi^2}{16}}$ equals (2007 - 3 marks)
 (a) $\frac{8}{\pi} f(2)$ (b) $\frac{2}{\pi} f(2)$ (c) $\frac{2}{\pi} f\left(\frac{1}{2}\right)$ (d) $4f(2)$
32. Let $g(x) = \frac{(x-1)^n}{\log \cos^m(x-1)}$; $0 < x < 2$, m and n are integers, $m \neq 0$, $n > 0$, and let p be the left hand derivative of $|x-1|$ at $x = 1$. If $\lim_{x \rightarrow 1^+} g(x) = p$, then (2008)
 (a) $n = 1, m = 1$ (b) $n = 1, m = -1$
 (c) $n = 2, m = 2$ (d) $n > 2, m = n$
33. If $\lim_{x \rightarrow 0} \left[1 + x \ln(1+b^2) \right]^{1/x} = 2b \sin^2 \theta$, $b > 0$ and $\theta \in (-\pi, \pi]$, then the value of θ is (2011)
 (a) $\pm \frac{\pi}{4}$ (b) $\pm \frac{\pi}{3}$ (c) $\pm \frac{\pi}{6}$ (d) $\pm \frac{\pi}{2}$
34. If $\lim_{x \rightarrow \infty} \left(\frac{x^2 + x + 1}{x + 1} - ax - b \right) = 4$, then (2012)
 (a) $a = 1, b = 4$ (b) $a = 1, b = -4$
 (c) $a = 2, b = -3$ (d) $a = 2, b = 3$
35. Let $f(x) = \begin{cases} x^2 \left| \cos \frac{\pi}{x} \right|, & x \neq 0 \\ 0, & x = 0 \end{cases}$, $x \in R$ then f is (2012)
 (a) differentiable both at $x = 0$ and at $x = 2$
 (b) differentiable at $x = 0$ but not differentiable at $x = 2$
 (c) not differentiable at $x = 0$ but differentiable at $x = 2$
 (d) differentiable neither at $x = 0$ nor at $x = 2$
36. Let $\alpha(a)$ and $\beta(a)$ be the roots of the equation $(\sqrt[3]{1+a} - 1)x^2 + (\sqrt{1+a} - 1)x + (\sqrt[6]{1+a} - 1) = 0$ where $a > -1$. Then $\lim_{a \rightarrow 0^+} \alpha(a)$ and $\lim_{a \rightarrow 0^+} \beta(a)$ are (2012)
 (a) $-\frac{5}{2}$ and 1 (b) $-\frac{1}{2}$ and -1
 (c) $-\frac{7}{2}$ and 2 (d) $-\frac{9}{2}$ and 3

D MCQs with One or More than One Correct

1. If $x + |y| = 2y$, then y as a function of x is (1984 - 3 Marks)
 (a) defined for all real x
 (b) continuous at $x = 0$
 (c) differentiable for all x
 (d) such that $\frac{dy}{dx} = \frac{1}{3}$ for $x < 0$
2. If $f(x) = x(\sqrt{x} - \sqrt{x+1})$, then— (1985 - 2 Marks)
 (a) $f(x)$ is continuous but not differentiable at $x = 0$
 (b) $f(x)$ is differentiable at $x = 0$
 (c) $f(x)$ is not differentiable at $x = 0$
 (d) none of these

3. The function $f(x) = 1 + |\sin x|$ is (1986 - 2 Marks)
 (a) continuous nowhere
 (b) continuous everywhere
 (c) differentiable nowhere
 (d) not differentiable at $x = 0$
 (e) not differentiable at infinite number of points.
4. Let $[x]$ denote the greatest integer less than or equal to x . If $f(x) = [x \sin \pi x]$, then $f(x)$ is (1986 - 2 Marks)
 (a) continuous at $x = 0$ (b) continuous in $(-1, 0)$
 (c) differentiable at $x = 1$ (d) differentiable in $(-1, 1)$
 (e) none of these
5. The set of all points where the function $f(x) = \frac{x}{(1 + |x|)}$ is differentiable, is (1987 - 2 Marks)
 (a) $(-\infty, \infty)$ (b) $[0, \infty)$
 (c) $(-\infty, 0) \cup (0, \infty)$ (d) $(0, \infty)$
 (e) None
6. The function $f(x) = \begin{cases} |x-3|, & x \geq 1 \\ \frac{x^2}{4} - \frac{3x}{2} + \frac{13}{4}, & x < 1 \end{cases}$ is (1988 - 2 Marks)
 (a) continuous at $x = 1$ (b) differentiable at $x = 1$
 (c) continuous at $x = 3$ (d) differentiable at $x = 3$.
7. If $f(x) = \frac{1}{2}x - 1$, then on the interval $[0, \pi]$ (1989 - 2 Marks)
 (a) $\tan [f(x)]$ and $1/f(x)$ are both continuous
 (b) $\tan [f(x)]$ and $1/f(x)$ are both discontinuous
 (c) $\tan [f(x)]$ and $f^{-1}(x)$ are both continuous
 (d) $\tan [f(x)]$ is continuous but $1/f(x)$ is not.
8. The value of $\lim_{x \rightarrow 0} \frac{\sqrt{\frac{1}{2}(1 - \cos 2x)}}{x}$ (1991 - 2 Marks)
 (a) 1 (b) -1
 (c) 0 (d) none of these
9. The following functions are continuous on $(0, \pi)$. (1991 - 2 Marks)
 (a) $\tan x$
 (b) $\int_0^x t \sin \frac{1}{t} dt$
 (c) $\begin{cases} 1, & 0 < x \leq \frac{3\pi}{4} \\ 2 \sin \frac{2}{9}x, & \frac{3\pi}{4} < x < \pi \end{cases}$
 (d) $\begin{cases} x \sin x, & 0 < x \leq \frac{\pi}{2} \\ \frac{\pi}{2} \sin(\pi + x), & \frac{\pi}{2} < x < \pi \end{cases}$
10. Let $f(x) = \begin{cases} 0, & x < 0 \\ x^2, & x \geq 0 \end{cases}$ then for all x (1994)
 (a) f' is differentiable (b) f is differentiable
 (c) f' is continuous (d) f is continuous
11. Let $g(x) = xf(x)$, where $f(x) = \begin{cases} x \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$. At $x = 0$
 (a) g is differentiable but g' is not continuous (1994)
 (b) g is differentiable while f is not
 (c) both f and g are differentiable
 (d) g is differentiable and g' is continuous
12. The function $f(x) = \max \{(1-x), (1+x), 2\}$, $x \in (-\infty, \infty)$ is (1995)
 (a) continuous at all points
 (b) differentiable at all points
 (c) differentiable at all points except at $x = 1$ and $x = -1$
 (d) continuous at all points except at $x = 1$ and $x = -1$, where it is discontinuous
13. Let $h(x) = \min \{x, x^2\}$, for every real number of x , Then (1998 - 2 Marks)
 (a) h is continuous for all x
 (b) h is differentiable for all x
 (c) $h'(x) = 1$, for all $x > 1$
 (d) h is not differentiable at two values of x .
14. $\lim_{x \rightarrow 1} \frac{\sqrt{1 - \cos 2(x-1)}}{x-1}$ (1998 - 2 Marks)
 (a) exists and it equals $\sqrt{2}$
 (b) exists and it equals $-\sqrt{2}$
 (c) does not exist because $x-1 \rightarrow 0$
 (d) does not exist because the left hand limit is not equal to the right hand limit.
15. If $f(x) = \min \{1, x^2, x^3\}$, then (2006 - 5M, -1)
 (a) $f(x)$ is continuous $\forall x \in \mathbb{R}$
 (b) $f(x)$ is continuous and differentiable everywhere.
 (c) $f(x)$ is not differentiable at two points
 (d) $f(x)$ is not differentiable at one point
16. Let $L = \lim_{x \rightarrow 0} \frac{a - \sqrt{a^2 - x^2} - \frac{x^2}{4}}{x^4}$, $a > 0$.
 If L is finite, then (2009)
 (a) $a = 2$ (b) $a = 1$ (c) $L = \frac{1}{64}$ (d) $L = \frac{1}{32}$
17. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x+y) = f(x) + f(y)$, $\forall x, y \in \mathbb{R}$. If $f(x)$ is differentiable at $x = 0$, then (2011)
 (a) $f(x)$ is differentiable only in a finite interval containing zero
 (b) $f(x)$ is continuous $\forall x \in \mathbb{R}$
 (c) $f'(x)$ is constant $\forall x \in \mathbb{R}$
 (d) $f(x)$ is differentiable except at finitely many points.

18. If $f(x) = \begin{cases} -x - \frac{\pi}{2}, & x \leq -\frac{\pi}{2} \\ -\cos x, & -\frac{\pi}{2} < x \leq 0, \\ x-1, & 0 < x \leq 1 \\ \ln x, & x > 1 \end{cases}$ then (2011)

- (a) $f(x)$ is continuous at $x = -\frac{\pi}{2}$
 (b) $f(x)$ is not differentiable at $x = 0$
 (c) $f(x)$ is differentiable at $x = 1$
 (d) $f(x)$ is differentiable at $x = -\frac{3}{2}$

19. For every integer n , let a_n and b_n be real numbers. Let function $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by (2012)

$$f(x) = \begin{cases} a_n + \sin \pi x, & \text{for } x \in [2n, 2n+1] \\ b_n + \cos \pi x, & \text{for } x \in (2n-1, 2n) \end{cases}$$

for all integers n . If f is continuous, then which of the following hold(s) for all n ?

- (a) $a_{n-1} - b_{n-1} = 0$ (b) $a_n - b_n = 1$
 (c) $a_n - b_{n+1} = 1$ (d) $a_{n-1} - b_n = -1$

20. For $a \in \mathbb{R}$ (the set of all real numbers), $a \neq -1$, (JEE Adv. 2013)

$$\lim_{n \rightarrow \infty} \frac{(1^a + 2^a + \dots + n^a)}{(n+1)^{a-1}[(na+1) + (na+2) + \dots + (na+n)]} = \frac{1}{60}.$$

Then $a =$

- (a) 5 (b) 7 (c) $\frac{-15}{2}$ (d) $\frac{-17}{2}$

21. Let $f: [a, b] \rightarrow [1, \infty)$ be a continuous function and let $g: \mathbb{R} \rightarrow \mathbb{R}$ be defined as (JEE Adv. 2014)

$$g(x) = \begin{cases} 0, & \text{if } x < a, \\ \int_a^x f(t) dt, & \text{if } a \leq x \leq b; \\ \int_a^b f(t) dt, & \text{if } x > b. \end{cases}$$

- (a) $g(x)$ is continuous but not differentiable at a
 (b) $g(x)$ is differentiable on $\mathbb{R}$
 (c) $g(x)$ is continuous but not differentiable at b
 (d) $g(x)$ is continuous and differentiable at either (a) or (b) but not both

22. For every pair of continuous functions $f, g: [0, 1] \rightarrow \mathbb{R}$ such that $\max \{f(x): x \in [0, 1]\} = \max \{g(x): x \in [0, 1]\}$, the correct statement(s) is (are): (JEE Adv. 2014)

- (a) $(f(c))^2 + 3f(c) = (g(c))^2 + 3g(c)$ for some $c \in [0, 1]$
 (b) $(f(c))^2 + f(c) = (g(c))^2 + 3g(c)$ for some $c \in [0, 1]$
 (c) $(f(c))^2 + 3f(c) = (g(c))^2 + g(c)$ for some $c \in [0, 1]$
 (d) $(f(c))^2 = (g(c))^2$ for some $c \in [0, 1]$

23. Let $g: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function with $g(0) = 0$,

$$g'(0) = 0 \text{ and } g'(1) \neq 0. \text{ Let } f(x) = \begin{cases} \frac{x}{|x|} g(x), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

and $h(x) = e^{|x|}$ for all $x \in \mathbb{R}$. Let $(f \circ h)(x)$ denote $f(h(x))$ and $(h \circ f)(x)$ denote $h(f(x))$. Then which of the following is (are) true?

(JEE Adv. 2015)

- (a) f is differentiable at $x = 0$
 (b) h is differentiable at $x = 0$
 (c) $f \circ h$ is differentiable at $x = 0$
 (d) $h \circ f$ is differentiable at $x = 0$

24. Let $a, b \in \mathbb{R}$ and $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = a \cos(|x^3 - x|) + b|x| \sin(|x^3 + x|)$.

Then f is (JEE Adv. 2016)

- (a) differentiable at $x=0$ if $a=0$ and $b=1$
 (b) differentiable at $x=1$ if $a=1$ and $b=0$
 (c) NOT differentiable at $x=0$ if $a=1$ $b=0$
 (d) NOT differentiable at $x=1$ if $a=1$ and $b=1$

25. Let $f: \left[-\frac{1}{2}, 2\right] \rightarrow \mathbb{R}$ and $g: \left[-\frac{1}{2}, 2\right] \rightarrow \mathbb{R}$ be functions defined by $f(x) = [x^2 - 3]$ and $g(x) = |x|f(x) + |4x - 7|f(x)$, where $[y]$ denotes the greatest integer less than or equal to y for $y \in \mathbb{R}$. Then (JEE Adv. 2016)

- (a) f is discontinuous exactly at three points in $\left[-\frac{1}{2}, 2\right]$
 (b) f is discontinuous exactly at four points in $\left[-\frac{1}{2}, 2\right]$
 (c) g is NOT differentiable exactly at four points in $\left(-\frac{1}{2}, 2\right)$
 (d) g is NOT differentiable exactly at five points in $\left(-\frac{1}{2}, 2\right)$

E Subjective Problems

1. Evaluate $\lim_{x \rightarrow a} \frac{\sqrt{a+2x} - \sqrt{3x}}{\sqrt{3a+x} - 2\sqrt{x}}$, ($a \neq 0$) (1978)
 2. $f(x)$ is the integral of $\frac{2 \sin x - \sin 2x}{x^3}$, $x \neq 0$, find $\lim_{x \rightarrow 0} f'(x)$ (1979)

3. Evaluate: $\lim_{h \rightarrow 0} \frac{(a+h)^2 \sin(a+h) - a^2 \sin a}{h}$ (1980)

4. Let $f(x+y) = f(x) + f(y)$ for all x and y . If the function $f(x)$ is continuous at $x = 0$, then show that $f(x)$ is continuous at all x . (1981 - 2 Marks)

5. Use the formula $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ln a$ to find

$$\lim_{x \rightarrow 0} \frac{2^x - 1}{(1+x)^{1/2} - 1} \quad (1982 - 2 \text{ Marks})$$

6. Let $f(x) = \begin{cases} 1+x, & 0 \leq x \leq 2 \\ 3-x, & 2 \leq x \leq 3 \end{cases}$ (1983 - 2 Marks)

Determine the form of $g(x) = f[f(x)]$ and hence find the points of discontinuity of g , if any

7. Let $f(x) = \begin{cases} \frac{x^2}{2}, & 0 \leq x < 1 \\ 2x^2 - 3x + \frac{3}{2}, & 1 \leq x \leq 2 \end{cases}$ (1983 - 2 Marks)

Discuss the continuity of f, f' and f'' on $[0, 2]$.

8. Let $f(x) = x^3 - x^2 + x + 1$ and $g(x) = \max\{f(t); 0 \leq t \leq x\}, 0 \leq x \leq 1$ (1985 - 5 Marks)
 $= 3 - x \quad 0 \leq x \leq 2$

Discuss the continuity and differentiability of the function $g(x)$ in the interval $(0, 2)$.

9. Let $f(x)$ be defined in the interval $[-2, 2]$ such that

$$f(x) = \begin{cases} -1, & -2 \leq x \leq 0 \\ x-1, & 0 < x \leq 2 \end{cases}$$

and $g(x) = f(|x|) + |f(x)|$

Test the differentiability of $g(x)$ in $(-2, 2)$. (1986 - 5 Marks)

10. Let $f(x)$ be a continuous and $g(x)$ be a discontinuous function. prove that $f(x) + g(x)$ is a discontinuous function. (1987 - 2 Marks)

11. Let $f(x)$ be a function satisfying the condition $f(-x) = f(x)$ for all real x . If $f'(0)$ exists, find its value. (1987 - 2 Marks)

12. Find the values of a and b so that the function

$$f(x) = \begin{cases} x + a\sqrt{2} \sin x, & 0 \leq x < \pi/4 \\ 2x \cot x + b, & \pi/4 \leq x \leq \pi/2 \\ a \cos 2x - b \sin x, & \pi/2 < x \leq \pi \end{cases}$$

is continuous for $0 \leq x \leq \pi$. (1989 - 2 Marks)

13. Draw a graph of the function $y = [x] + |1-x|, -1 \leq x \leq 3$. Determine the points, if any, where this function is not differentiable. (1989 - 4 Marks)

14. Let $f(x) = \begin{cases} \frac{1 - \cos 4x}{x^2}, & x < 0 \\ a, & x = 0 \\ \frac{\sqrt{x}}{\sqrt{16 + \sqrt{x}} - 4}, & x > 0 \end{cases}$ (1990 - 4 Marks)

Determine the value of a , if possible, so that the function is continuous at $x = 0$

15. A function $f: R \rightarrow R$ satisfies the equation $f(x+y) = f(x)f(y)$ for all x, y in R and $f(x) \neq 0$ for any x in R . Let the function be differentiable at $x = 0$ and $f'(0) = 2$. Show that $f'(x) = 2f(x)$ for all x in R . Hence, determine $f(x)$. (1990 - 4 Marks)

16. Find $\lim_{x \rightarrow 0} \{\tan(\pi/4 + x)\}^{1/x}$ (1993 - 2 Marks)

17. Let $f(x) = \begin{cases} \{1 + |\sin x|\}^{a/|\sin x|}; & \frac{\pi}{6} < x < 0 \\ b & x = 0 \\ e^{\tan 2x / \tan 3x} & 0 < x < \frac{\pi}{6} \end{cases}$ (1994 - 4 Marks)

Determine a and b such that $f(x)$ is continuous at $x = 0$

18. Let $f\left(\frac{x+y}{2}\right) = \frac{f(x) + f(y)}{2}$ for all real x and y . If $f'(0)$ exists and equals -1 and $f(0) = 1$, find $f(2)$. (1995 - 5 Marks)

19. Determine the values of x for which the following function fails to be continuous or differentiable: (1997 - 5 Marks)

$$f(x) = \begin{cases} 1-x, & x < 1 \\ (1-x)(2-x), & 1 \leq x \leq 2 \\ 3-x, & x > 2 \end{cases} \text{ Justify your answer.}$$

20. Let $f(x), x \geq 0$, be a non-negative continuous function, and

$$\text{let } F(x) = \int_0^x f(t) dt, x \geq 0. \text{ If for some } c > 0, f(x) \leq cF(x) \text{ for all}$$

$x \geq 0$, then show that $f(x) = 0$ for all $x \geq 0$. (2001 - 5 Marks)

21. Let $\alpha \in R$. Prove that a function $f: R \rightarrow R$ is differentiable at α if and only if there is a function $g: R \rightarrow R$ which is continuous at α and satisfies $f(x) - f(\alpha) = g(x)(x - \alpha)$ for all $x \in R$. (2001 - 5 Marks)

22. Let $f(x) = \begin{cases} x+a & \text{if } x < 0 \\ |x-1| & \text{if } x \geq 0, \end{cases}$ and (2002 - 5 Marks)

$$g(x) = \begin{cases} x+1 & \text{if } x < 0 \\ (x-1)^2 + b & \text{if } x \geq 0, \end{cases} \text{ where } a \text{ and } b \text{ are}$$

non-negative real numbers. Determine the composite function $g \circ f$. If $(g \circ f)(x)$ is continuous for all real x , determine the values of a and b . Further, for these values of a and b , is $g \circ f$ differentiable at $x = 0$? Justify your answer.

Limits, Continuity and Differentiability

23. If a function $f : [-2a, 2a] \rightarrow R$ is an odd function such that $f(x) = f(2a - x)$ for $x \in [a, 2a]$ and the left hand derivative at $x = a$ is 0 then find the left hand derivative at $x = -a$.

(2003 - 2 Marks)

24. $f'(0) = \lim_{n \rightarrow \infty} n f\left(\frac{1}{n}\right)$ and $f(0) = 0$. Using this find

$$\lim_{n \rightarrow \infty} \left((n+1) \frac{2}{\pi} \cos^{-1}\left(\frac{1}{n}\right) - n \right), \left| \cos^{-1} \frac{1}{n} \right| < \frac{\pi}{2}$$

(2004 - 2 Marks)

25. If $|c| \leq \frac{1}{2}$ and $f(x)$ is a differentiable function at $x = 0$ given

$$\text{by } f(x) = \begin{cases} b \sin^{-1}\left(\frac{c+x}{2}\right), & -\frac{1}{2} < x < 0 \\ \frac{1}{2}, & x = 0 \\ \frac{e^{ax/2} - 1}{x}, & 0 < x < \frac{1}{2} \end{cases}$$

Find the value of 'a' and prove that $64b^2 = 4 - c^2$

(2004 - 4 Marks)

26. If $f(x-y) = f(x) \cdot g(y) - f(y) \cdot g(x)$ and

$$g(x-y) = g(x) \cdot g(y) - f(x) \cdot f(y) \text{ for all } x, y \in R.$$

If right hand derivative at $x = 0$ exists for $f(x)$. Find derivative of $g(x)$ at $x = 0$

(2005 - 4 Marks)

F Integer Value Correct Type

DIRECTIONS (Q. 1 and 2) : Each question contains statements given in two columns, which have to be matched. The statements in Column-I are labelled A, B, C and D, while the statements in Column-II are labelled p, q, r, s and t. Any given statement in Column-I can have correct matching with ONE OR MORE statement(s) in Column-II. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example :

If the correct matches are A-p, s and t; B-q and r; C-p and q; and D-s then the correct darkening of bubbles will look like the given.

	p	q	r	s	t
A	☒	☒	☐	☒	☒
B	☐	☒	☒	☐	☐
C	☒	☒	☐	☐	☐
D	☐	☐	☐	☒	☐

1. In this questions there are entries in columns I and II. Each entry in column I is related to exactly one entry in column II. Write the correct letter from column II against the entry number in column I in your answer book.

(1992 - 2 Marks)

Column I

- (A) $\sin(\pi[x])$
(B) $\sin(\pi(x-[x]))$

Column II

- (p) differentiable everywhere
(q) nowhere differentiable
(r) not differentiable at 1 and -1

2. In the following $[x]$ denotes the greatest integer less than or equal to x .

Match the functions in **Column I** with the properties in **Column II** and indicate your answer by darkening the appropriate bubbles in the 4×4 matrix given in the ORS.

(2007 - 6 marks)

Column I

- (A) $x|x|$
(B) $\sqrt{|x|}$
(C) $x + [x]$
(D) $|x-1| + |x+1|$

Column II

- (p) continuous in $(-1, 1)$
(q) differentiable in $(-1, 1)$
(r) strictly increasing in $(-1, 1)$
(s) not differentiable at least at one point in $(-1, 1)$

DIRECTIONS (Q. 3) : Following question has matching lists. The codes for the list have choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

3. Let $f_1 : R \rightarrow R$, $f_2 : [0, \infty) \rightarrow R$, $f_3 : R \rightarrow R$ and $f_4 : R \rightarrow [0, \infty)$ be defined by $f_1(x) = \begin{cases} |x| & \text{if } x < 0, \\ e^x & \text{if } x \geq 0; \end{cases}$

$$f_2(x) = x^2; f_3(x) = \begin{cases} \sin x & \text{if } x < 0, \\ x & \text{if } x \geq 0; \end{cases} \text{ and } f_4(x) = \begin{cases} f_2(f_1(x)) & \text{if } x < 0, \\ f_2(f_1(x)) - 1 & \text{if } x \geq 0. \end{cases}$$

(JEE Adv. 2014)

List-I

- P. f_4 is
 Q. f_3 is
 R. $f_2 \circ f_1$ is
 S. f_2 is

	P	Q	R	S
(a)	3	1	4	2
(c)	3	1	2	4

I Integer Value Correct Type

- Let $f: [1, \infty) \rightarrow [2, \infty)$ be a differentiable function such that $f(1) = 2$. If $\int_1^x f(t) dt = 3xf(x) - x^3$ for all $x \geq 1$, then the value of $f(2)$ is (2011)
- The largest value of non-negative integer a for which $\lim_{x \rightarrow 1} \left\{ \frac{-ax + \sin(x-1) + a}{x + \sin(x-1) - 1} \right\}^{\frac{1-x}{1-\sqrt{x}}} = \frac{1}{4}$ is (JEE Adv. 2014)
- Let $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be respectively given by $f(x) = |x| + 1$ and $g(x) = x^2 + 1$. Define $h: \mathbb{R} \rightarrow \mathbb{R}$ by

List-II

- Onto but not one-one
- Neither continuous nor one-one
- Differentiable but not one-one
- Continuous and one-one

	P	Q	R	S
(b)	1	3	4	2
(d)	1	3	2	4

$$h(x) = \begin{cases} \max \{f(x), g(x)\} & \text{if } x \leq 0, \\ \min \{f(x), g(x)\} & \text{if } x > 0. \end{cases}$$

The number of points at which $h(x)$ is not differentiable is (JEE Adv. 2014)

- Let m and n be two positive integers greater than 1. If $\lim_{\alpha \rightarrow 0} \left(\frac{e^{\cos(\alpha^n)} - e}{\alpha^m} \right) = -\left(\frac{e}{2}\right)$ then the value of $\frac{m}{n}$ is (JEE Adv. 2015)
- Let $\alpha, \beta \in \mathbb{R}$ be such that $\lim_{x \rightarrow 0} \frac{x^2 \sin(\beta x)}{\alpha x - \sin x} = 1$. Then $6(\alpha + \beta)$ equals. (JEE Adv. 2016)

Section-B

JEE Main / AIEEE

- $\lim_{x \rightarrow 0} \frac{\sqrt{1 - \cos 2x}}{\sqrt{2x}}$ is [2002]

(a) 1	(b) -1
(c) zero	(d) does not exist
- $\lim_{x \rightarrow \infty} \left(\frac{x^2 + 5x + 3}{x^2 + x + 3} \right)^x$ [2002]

(a) e^4	(b) e^2	(c) e^3	(d) 1
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- Let $f(x) = 4$ and $f'(x) = 4$. Then $\lim_{x \rightarrow 2} \frac{xf(2) - 2f(x)}{x-2}$ is given by [2002]

(a) 2	(b) -2	(c) -4	(d) 3
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- $\lim_{n \rightarrow \infty} \frac{1^p + 2^p + 3^p + \dots + n^p}{n^{p+1}}$ is [2002]

(a) $\frac{1}{p+1}$	(b) $\frac{1}{1-p}$	(c) $\frac{1}{p} - \frac{1}{p-1}$	(d) $\frac{1}{p+2}$
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- $\lim_{x \rightarrow 0} \frac{\log x^n - [x]}{[x]}$, $n \in \mathbb{N}$, ($[x]$ denotes greatest integer less than or equal to x) [2002]

(a) has value -1	(b) has value 0
(c) has value 1	(d) does not exist
- If $f(1) = 1, f'(1) = 2$, then $\lim_{x \rightarrow 1} \frac{\sqrt{f(x)} - 1}{\sqrt{x} - 1}$ is [2002]

(a) 2	(b) 4	(c) 1	(d) 1/2
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- f is defined in $[-5, 5]$ as [2002]

$$f(x) = x \text{ if } x \text{ is rational}$$

$$= -x \text{ if } x \text{ is irrational. Then}$$

(a) $f(x)$ is continuous at every x , except $x = 0$
(b) $f(x)$ is discontinuous at every x , except $x = 0$
(c) $f(x)$ is continuous everywhere
(d) $f(x)$ is discontinuous everywhere
- $f(x)$ and $g(x)$ are two differentiable functions on $[0, 2]$ such that $f''(x) - g''(x) = 0$, $f'(1) = 2g'(1) = 4f(2) = 3g(2) = 9$ then $f(x) - g(x)$ at $x = 3/2$ is [2002]

(a) 0	(b) 2	(c) 10	(d) 5
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Limits, Continuity and Differentiability

9. If $f(x+y) = f(x) \cdot f(y) \forall x, y$ and $f(5) = 2, f'(0) = 3$, then $f'(5)$ is [2002]
 (a) 0 (b) 1 (c) 6 (d) 2
10. $\lim_{n \rightarrow \infty} \frac{1+2^4+3^4+\dots+n^4}{n^5} - \lim_{n \rightarrow \infty} \frac{1+2^3+3^3+\dots+n^3}{n^5}$ [2003]
 (a) $\frac{1}{5}$ (b) $\frac{1}{30}$ (c) Zero (d) $\frac{1}{4}$
11. If $\lim_{x \rightarrow 0} \frac{\log(3+x) - \log(3-x)}{x} = k$, the value of k is [2003]
 (a) $-\frac{2}{3}$ (b) 0 (c) $-\frac{1}{3}$ (d) $\frac{2}{3}$
12. The value of $\lim_{x \rightarrow 0} \frac{\int_0^{x^2} \sec^2 t dt}{x \sin x}$ is [2003]
 (a) 0 (b) 3 (c) 2 (d) 1
13. Let $f(a) = g(a) = k$ and their n th derivatives $f^n(a), g^n(a)$ exist and are not equal for some n . Further if $\lim_{x \rightarrow a} \frac{f(a)g(x) - f(a) - g(a)f(x) + f(a)}{g(x) - f(x)} = 4$ then the value of k is [2003]
 (a) 0 (b) 4 (c) 2 (d) 1
14. $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\left[1 - \tan\left(\frac{x}{2}\right)\right][1 - \sin x]}{\left[1 + \tan\left(\frac{x}{2}\right)\right][\pi - 2x]^3}$ is [2003]
 (a) ∞ (b) $\frac{1}{8}$ (c) 0 (d) $\frac{1}{32}$
15. If $f(x) = \begin{cases} xe^{-\left(\frac{1}{|x|} + \frac{1}{x}\right)}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ then $f(x)$ is [2003]
 (a) discontinuous every where
 (b) continuous as well as differentiable for all x
 (c) continuous for all x but not differentiable at $x = 0$
 (d) neither differentiable nor continuous at $x = 0$
16. If $\lim_{x \rightarrow \infty} \left(1 + \frac{a}{x} + \frac{b}{x^2}\right)^{2x} = e^2$, then the values of a and b , are [2004]
 (a) $a = 1$ and $b = 2$ (b) $a = 1, b \in \mathbb{R}$
 (c) $a \in \mathbb{R}, b = 2$ (d) $a \in \mathbb{R}, b \in \mathbb{R}$
17. Let $f(x) = \frac{1 - \tan x}{4x - \pi}, x \neq \frac{\pi}{4}, x \in \left[0, \frac{\pi}{2}\right]$. If $f(x)$ is continuous in $\left[0, \frac{\pi}{2}\right]$, then $f\left(\frac{\pi}{4}\right)$ is [2004]
 (a) -1 (b) $\frac{1}{2}$ (c) $-\frac{1}{2}$ (d) 1
18. $\lim_{n \rightarrow \infty} \left[\frac{1}{n^2} \sec^2 \frac{1}{n^2} + \frac{2}{n^2} \sec^2 \frac{4}{n^2} + \dots + \frac{1}{n} \sec^2 1 \right]$ equals [2005]
 (a) $\frac{1}{2} \sec 1$ (b) $\frac{1}{2} \operatorname{cosec} 1$
 (c) $\tan 1$ (d) $\frac{1}{2} \tan 1$
19. Let α and β be the distinct roots of $ax^2 + bx + c = 0$, then $\lim_{x \rightarrow \alpha} \frac{1 - \cos(ax^2 + bx + c)}{(x - \alpha)^2}$ is equal to [2005]
 (a) $\frac{a^2}{2}(\alpha - \beta)^2$ (b) 0
 (c) $\frac{-a^2}{2}(\alpha - \beta)^2$ (d) $\frac{1}{2}(\alpha - \beta)^2$
20. Suppose $f(x)$ is differentiable at $x = 1$ and $\lim_{h \rightarrow 0} \frac{1}{h} f(1+h) = 5$, then $f'(1)$ equals [2005]
 (a) 3 (b) 4 (c) 5 (d) 6
21. Let f be differentiable for all x . If $f(1) = -2$ and $f'(x) \geq 2$ for $x \in [1, 6]$, then [2005]
 (a) $f(6) \geq 8$ (b) $f(6) < 8$ (c) $f(6) < 5$ (d) $f(6) = 5$
22. If f is a real valued differentiable function satisfying $|f(x) - f(y)| \leq (x - y)^2, x, y \in \mathbb{R}$ and $f(0) = 0$, then $f(1)$ equals [2005]
 (a) -1 (b) 0 (c) 2 (d) 1
23. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \min \{x+1, |x|+1\}$, Then which of the following is true?
 (a) $f(x)$ is differentiable everywhere [2007]
 (b) $f(x)$ is not differentiable at $x = 0$
 (c) $f(x) \geq 1$ for all $x \in \mathbb{R}$
 (d) $f(x)$ is not differentiable at $x = 1$
24. The function $f: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ given by [2007]

$$f(x) = \frac{1}{x} - \frac{2}{e^{2x} - 1}$$
 can be made continuous at $x = 0$ by defining $f(0)$ as
 (a) 0 (b) 1 (c) 2 (d) -1

25. Let $f(x) = \begin{cases} (x-1)\sin \frac{1}{x-1} & \text{if } x \neq 1 \\ 0 & \text{if } x = 1 \end{cases}$ [2008]

Then which one of the following is true?

- (a) f is neither differentiable at $x=0$ nor at $x=1$
 (b) f is differentiable at $x=0$ and at $x=1$
 (c) f is differentiable at $x=0$ but not at $x=1$
 (d) f is differentiable at $x=1$ but not at $x=0$

26. Let $f: \mathbf{R} \rightarrow \mathbf{R}$ be a positive increasing function with

$\lim_{x \rightarrow \infty} \frac{f(3x)}{f(x)} = 1$. Then $\lim_{x \rightarrow \infty} \frac{f(2x)}{f(x)} =$ [2010]

- (a) $\frac{2}{3}$ (b) $\frac{3}{2}$ (c) 3 (d) 1

27. $\lim_{x \rightarrow 2} \left(\frac{\sqrt{1 - \cos\{2(x-2)\}}}{x-2} \right)$ [2011]

- (a) equals $\sqrt{2}$ (b) equals $-\sqrt{2}$
 (c) equals $\frac{1}{\sqrt{2}}$ (d) does not exist

28. The values of p and q for which the function [2011]

$$f(x) = \begin{cases} \frac{\sin(p+1)x + \sin x}{x}, & x < 0 \\ q, & x = 0 \\ \frac{\sqrt{x+x^2} - \sqrt{x}}{x^{3/2}}, & x > 0 \end{cases}$$

$x=0$ is continuous for all x in $\mathbf{R}$, are

- (a) $p = \frac{5}{2}, q = \frac{1}{2}$ (b) $p = -\frac{3}{2}, q = \frac{1}{2}$
 (c) $p = \frac{1}{2}, q = \frac{3}{2}$ (d) $p = \frac{1}{2}, q = -\frac{3}{2}$

29. Let $f: \mathbf{R} \rightarrow [0, \infty)$ be such that $\lim_{x \rightarrow 5} f(x)$ exists and

$\lim_{x \rightarrow 5} \frac{(f(x))^2 - 9}{\sqrt{|x-5|}} = 0$. Then $\lim_{x \rightarrow 5} f(x)$ equals:

- (a) 0 (b) 1 (c) 2 (d) 3

30. If $f: \mathbf{R} \rightarrow \mathbf{R}$ is a function defined by $f(x) = [x]$

$\cos \left(\frac{2x-1}{2} \right) \pi$, where $[x]$ denotes the greatest integer function, then f is. [2012]

- (a) continuous for every real x .
 (b) discontinuous only at $x=0$
 (c) discontinuous only at non-zero integral values of x .
 (d) continuous only at $x=0$.

31. Consider the function, $f(x) = |x-2| + |x-5|, x \in \mathbf{R}$.

Statement-1: $f'(4) = 0$

Statement-2: f is continuous in $[2, 5]$, differentiable in $(2, 5)$ and $f(2) = f(5)$. [2012]

- (a) Statement-1 is false, Statement-2 is true.
 (b) Statement-1 is true, statement-2 is true; statement-2 is a correct explanation for Statement-1.
 (c) Statement-1 is true, statement-2 is true; statement-2 is **not** a correct explanation for Statement-1.
 (d) Statement-1 is true, statement-2 is false.

32. $\lim_{x \rightarrow 0} \frac{(1 - \cos 2x)(3 + \cos x)}{x \tan 4x}$ is equal to [JEE M 2013]

- (a) $-\frac{1}{4}$ (b) $\frac{1}{2}$ (c) 1 (d) 2

33. $\lim_{x \rightarrow 0} \frac{\sin(\pi \cos^2 x)}{x^2}$ is equal to: [JEE M 2014]

- (a) $-\pi$ (b) π (c) $\frac{\pi}{2}$ (d) 1

34. $\lim_{x \rightarrow 0} \frac{(1 - \cos 2x)(3 + \cos x)}{x \tan 4x}$ is equal to: [JEE M 2015]

- (a) 2 (b) $\frac{1}{2}$ (c) 4 (d) 3

35. If the function.

$g(x) = \begin{cases} k\sqrt{x+1}, & 0 \leq x \leq 3 \\ mx+2, & 3 < x \leq 5 \end{cases}$ is differentiable, then the value of $k+m$ is: [JEE M 2015]

- (a) $\frac{10}{3}$ (b) 4 (c) 2 (d) $\frac{16}{5}$

36. For $x \in \mathbf{R}$, $f(x) = |\log 2 - \sin x|$ and $g(x) = f(f(x))$, then:

[JEE M 2016]

- (a) $g'(0) = -\cos(\log 2)$
 (b) g is differentiable at $x=0$ and $g'(0) = -\sin(\log 2)$
 (c) g is not differentiable at $x=0$
 (d) $g'(0) = \cos(\log 2)$

37. $\lim_{n \rightarrow \infty} \left(\frac{(n+1)(n+2)\dots 3n}{n^{2n}} \right)^{\frac{1}{n}}$ is equal to: [JEE M 2016]

- (a) $\frac{9}{e^2}$ (b) $3 \log 3 - 2$
 (c) $\frac{18}{e^4}$ (d) $\frac{27}{e^2}$

38. Let $p = \lim_{x \rightarrow 0^+} \left(1 + \tan^2 \sqrt{x} \right)^{\frac{1}{2x}}$ then $\log p$ is equal to:

[JEE M 2016]

- (a) $\frac{1}{2}$ (b) $\frac{1}{4}$
 (c) 2 (d) 1



Limits, Continuity and Differentiability

Section-A : JEE Advanced/ IIT-JEE

- A** 1. $\{0\}$ 2. $k=7$ 3. $f(x) = \sqrt{4-x^2}, -2 \leq x \leq 0 = -\sqrt{4-x^2}, 0 \leq x \leq 2$
4. $\frac{2}{\pi}$ 5. 1 6. -1 7. 4 8. $\sqrt{2rh-h^2}, \frac{1}{128r}$ 9. e^5
10. $R - \{0\}$ 11. $(-\infty, -1) \cup [0, \infty)$, $I - \{0\}$ where I is the set of integer except $n = -1$
12. e^2 13. 10
- B** 1. F
- C** 1. (c) 2. (d) 3. (a) 4. (d) 5. (c) 6. (b) 7. (b)
8. (d) 9. (a) 10. (b) 11. (c) 12. (b) 13. (d) 14. (d)
15. (c) 16. (c) 17. (b) 18. (a) 19. (d) 20. (d) 21. (d)
22. (c) 23. (c) 24. (d) 25. (d) 26. (c) 27. (a) 28. (b)
29. (c) 30. (a) 31. (a) 32. (c) 33. (d) 34. (b) 35. (b)
36. (b)
- D** 1. (a, b, d) 2. (b) 3. (b, d, e) 4. (a, b, d) 5. (a) 6. (a, b, c)
7. (b) 8. (d) 9. (b, c) 10. (b, c, d) 11. (a, b) 12. (a, c)
13. (a, c, d) 14. (d) 15. (a, d) 16. (a, c) 17. (b, c) 18. (a, b, c, d)
19. (b, d) 20. (b, d) 21. (a, c) 22. (a, d) 23. (a, d) 24. (a, b)
25. (b, c)
- E** 1. $\frac{2}{3\sqrt{3}}$ 2. 1 3. $a^2 \cos a + 2a \sin a$ 5. $2 \ln 2$
6. $g(x) = \begin{cases} 2+x, & 0 \leq x \leq 1 \\ 2-x, & 1 < x \leq 2 \\ 4-x, & 2 < x \leq 3 \end{cases}$ discontinuity at $x = 1, 2$ 7. f and f' are continuous and f'' is discontinuous on $[0, 2]$
8. cont. on $(0, 2)$ and differentiable on $(0, 2) - \{1\}$ 9. not differentiable at $x = 1$ 11. $f'(0) = 0$
12. $a = \frac{\pi}{6}, b = -\frac{\pi}{12}$ 13. $x=0, 1, 2, 3$ 14. $a=8$ 15. $f(x) = e^{2x}$
16. e^2 17. $a = \frac{2}{3}, b = e^{2/3}$ 18. $f(2) = -1$
19. f is continuous and differentiable at all points except at $x = 2$.
22. $g(f(x)) = \begin{cases} x+a+1. & \text{if } x < -a \\ (x+a-1)^2 + b & \text{if } a \leq x < 0 \\ x^2 + b & \text{if } 0 \leq x \leq 1 \\ (x-2)^2 + b & \text{if } x > 1 \end{cases}, a=1, b=0, \text{ gof is differentiable at } x=0$ 23. 0
24. $\frac{\pi-2}{\pi}$ 25. 1 26. 0
- F** 1. (A) - p, (B) - r 2. (A) - p, q, r; (B) - p, s; (C) - r, s; (D) - p, q 3. (d)
- I** 1. 6 2. 2 3. 3 4. 2 5. 7

Section-B : JEE Main/ AIEEE

1. (d)	2. (a)	3. (c)	4. (a)	5. (d)	6. (a)	7. (b)
8. (d)	9. (c)	10. (a)	11. (d)	12. (d)	13. (b)	14. (d)
15. (c)	16. (b)	17. (c)	18. (d)	19. (a)	20. (c)	21. (a)
22. (b)	23. (a)	24. (b)	25. (c)	26. (d)	27. (d)	28. (b)
29. (d)	30. (a)	31. (c)	32. (d)	33. (b)	34. (a)	35. (c)
36. (d)	37. (d)	38. (a)				

Section-A

JEE Advanced/ IIT-JEE

A. Fill in the Blanks

1. Given $f(x) = \begin{cases} (x-1)^2 \sin \frac{1}{x-1} - |x|, & x \neq 1 \\ -1, & x = 1 \end{cases}$
We know that $|x|$ is not differentiable at $x = 0$
 $\therefore (x-1)^2 \sin \frac{1}{x-1} - |x|$ is not differentiable at $x = 0$.

At all other values of x , $f(x)$ is differentiable.

$\therefore$ The req. set of points is $\{0\}$.

2. It will be continuous at $x = 2$ if

$$\lim_{x \rightarrow 2} f(x) = f(2) \Rightarrow \lim_{x \rightarrow 2} \frac{x^3 + x^2 - 16x + 20}{(x-2)^2} = k$$

$$\Rightarrow k = \lim_{x \rightarrow 2} \frac{(x-2)^2(x+5)}{(x-2)^2} = \lim_{x \rightarrow 2} (x+5) = 7$$

$$\therefore k = 7$$

3. $f(x) = \sqrt{4-x^2}, -2 \leq x \leq 0 = -\sqrt{4-x^2}, 0 \leq x \leq 2$

By choosing any arcs of circle $x^2 - y^2 = 4$, we can define a discontinuous function, one of which is

$$f(x) = \begin{cases} \sqrt{4-x^2}, & -2 \leq x \leq 0 \\ -\sqrt{4-x^2}, & 0 \leq x \leq 2 \end{cases}$$

4. KEY CONCEPT

(L' Hospital rule)

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

if $\frac{f(a)}{g(a)}$ is of the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$ or $0 \times \infty$.

$$\lim_{x \rightarrow 1} (1-x) \tan \frac{\pi x}{2} \quad [\text{form } 0 \times \infty]$$

$$= \lim_{x \rightarrow 1} \frac{1-x}{\cot(\pi x/2)} \quad \left[\text{Form } \frac{0}{0} \right]$$

$$= \lim_{x \rightarrow 1} \frac{-1}{-\frac{\pi}{2} \operatorname{cosec}^2 \left(\frac{\pi x}{2} \right)} \quad [\text{Applying L' Hospital's rule}]$$

$$= \frac{2}{\pi}$$

5. Given that,

$$f(x) = \sin x, x \neq \pi, n = 0, \pm 1, \pm 2, \dots = 2, \text{ otherwise}$$

$$\text{And } g(x) = x^2 + 1, x \neq 0, 2$$

$$= 4, x = 0 = 5, \quad x = 2$$

$$\text{Then } \lim_{x \rightarrow 0} g[f(x)] = \lim_{x \rightarrow 0} g(\sin x) \Rightarrow \lim_{x \rightarrow 0} (\sin^2 x + 1) = 1$$

$$6. \lim_{x \rightarrow -\infty} \left[\frac{x^4 \sin \left(\frac{1}{x} \right) + x^2}{(1 + |x|^3)} \right]$$

$$\begin{aligned} \text{Let } L &= \lim_{x \rightarrow -\infty} \frac{x^3}{1 + |x|^3} \left[x \sin \left(\frac{1}{x} \right) + \frac{1}{x} \right] \\ &= \lim_{x \rightarrow -\infty} \frac{x^3}{|x|^3} \left[\frac{1}{1 + \frac{1}{|x|^3}} \right] \left[x \sin \left(\frac{1}{x} \right) + \frac{1}{x} \right] \dots (1) \\ &= \lim_{x \rightarrow -\infty} \frac{x^3}{|x|^3} \cdot 1 = \lim_{x \rightarrow -\infty} \frac{x^3}{-x^3} = -1 \end{aligned}$$

7. Given that $f(9) = 9, f'(9) = 4$

Then,

$$\lim_{x \rightarrow 9} \frac{\sqrt{f(x)} - 3}{\sqrt{x} - 3} = \lim_{x \rightarrow 9} \frac{(\sqrt{f(x)} - 3)(\sqrt{f(x)} + 3)}{(\sqrt{x} - 3)(\sqrt{x} + 3)}$$

$$\begin{aligned} \lim_{x \rightarrow 9} \frac{\sqrt{x} + 3}{\sqrt{f(x)} + 3} &= \lim_{x \rightarrow 9} \frac{f(x) - 9}{x - 9} \cdot \left[\frac{3 + 3}{3 + 3} \right] \\ &= \lim_{x \rightarrow 9} \frac{f(x) - f(9)}{x - 9} \cdot 1 = f'(9) = 4 \end{aligned}$$

8. In ΔABC , $AB = AC$

$AD \perp BC$

(D is mid pt of BC)

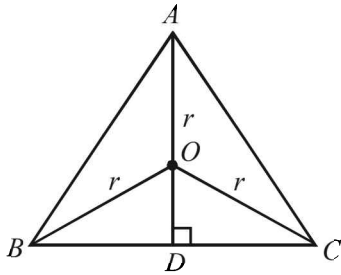
Let r = radius of circumcircle

$$\therefore OA = OB = OC = r$$

$$\text{Now } BD = \sqrt{BO^2 - OD^2} = \sqrt{r^2 - (h-r)^2}$$

$$= \sqrt{2rh - h^2}$$

$$\therefore BC = 2\sqrt{2rh - h^2}$$



$$\therefore \text{Area of } \triangle ABC = \frac{1}{2} \times BC \times AD = h\sqrt{2rh - h^2}$$

$$\begin{aligned} \text{Also } \lim_{h \rightarrow 0} \frac{A}{P^3} &= \lim_{h \rightarrow 0} \frac{h\sqrt{2rh - h^2}}{8(\sqrt{2rh - h^2} + \sqrt{2rh})^3} \\ &= \lim_{h \rightarrow 0} \frac{h^{3/2}\sqrt{2r - h}}{8h^{3/2}(\sqrt{2r - h} + \sqrt{2r})^3} \\ &= \lim_{h \rightarrow 0} \frac{\sqrt{2r - h}}{8(2r - h + \sqrt{2r})^3} \\ &= \frac{\sqrt{2r}}{8(\sqrt{2r} + \sqrt{2r})^3} = \frac{\sqrt{2r}}{8 \cdot 8 \cdot 2r \cdot \sqrt{2r}} = \frac{1}{128r} \end{aligned}$$

$$9. \lim_{x \rightarrow \infty} \left(\frac{x+6}{x+1} \right)^{x+4} = \lim_{x \rightarrow \infty} \left\{ \left[1 + \frac{5}{x+1} \right]^{\frac{x+1}{5}} \right\}^{5 \left(\frac{x+4}{x+1} \right)}$$

[Using $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^x = e$]

$$\lim_{x \rightarrow \infty} 5 \left(\frac{x+4}{x+1} \right) = e^{5 \lim_{x \rightarrow \infty} \left(\frac{1+4/x}{1+1/x} \right)} = e^5$$

10. We have,

$$f(x) = x|x| = \begin{cases} -x^2, & x < 0 \\ x^2, & x \geq 0 \end{cases}$$

$$f'(x) = \begin{cases} -2x, & x < 0 \\ 2x, & x \geq 0 \end{cases}$$

$$f''(x) = \begin{cases} -2, & x < 0 \\ 2, & x \geq 0 \end{cases}$$

Clearly $f''(x)$ exists at every pt. except at $x = 0$

Thus $f(x)$ is twice differentiable on $R - \{0\}$.

11. Thus function is not defined for those values of x for which $[x+1] = 0$. In other words it means that

$$0 \leq x+1 < 1 \text{ or } -1 \leq x < 0 \quad \dots\dots(1)$$

Hence the function is defined outside the region given by (1).

Required domain is $]-\infty, -1[\cup [0, \infty[$

Now, consider integral values of x say $x = n$

$$\text{R.H.L.} = \lim_{h \rightarrow 0} [n+h] \sin \frac{\pi}{[n+1+h]} = n \sin \frac{\pi}{(n+1)}$$

$$\text{L.H.L.} = \lim_{h \rightarrow 0} [n-h] \sin \frac{\pi}{[n+1-h]} = (n-1) \frac{\pi}{n}$$

Clearly $\text{RHL} \neq \text{LHL}$. Hence the given function is not continuous for integral values of n ($n \neq 0, -1$).

At $x = 0, f(0) = 0$,

$$\lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} [h] \sin \frac{\pi}{[h+1]} = 0$$

The function is not defined for $x < 0$. Hence we cannot find $\lim_{h \rightarrow 0} f(0-h)$. Thus $f(x)$ is continuous at $x = 0$. Hence the points of discontinuity are given by $I - \{0\}$ where I is set of integers n except $n = -1$

12. KEY CONCEPT

$$\lim_{x \rightarrow 0} [f(x)]^{g(x)} = e^{\lim_{x \rightarrow 0} g(x) \log f(x)}$$

$$\begin{aligned} \lim_{x \rightarrow 0} \left(\frac{1+5x^2}{1+3x^2} \right)^{1/x^2} &= e^{\lim_{x \rightarrow 0} \frac{1}{x^2} \log \left[\frac{1+5x^2}{1+3x^2} \right]} \\ &= e^{\lim_{x \rightarrow 0} \left[5 \cdot \frac{\log(1+5x^2)}{5x^2} - 3 \cdot \frac{\log(1+3x^2)}{3x^2} \right]} \\ &= e^{5-3} = e^2 \end{aligned}$$

13. Since $f(x)$ is given continuous on the closed bounded interval $[1, 3]$, $f(x)$ is bounded and assumes all the values lying in the interval $[m, M]$ where

$$m = \min f(x) \text{ and } M = \max f(x)$$

$$1 \leq x \leq 3 \Rightarrow f(1) \leq f(x) \leq f(3)$$

If $m < M$, then $f(x)$ must assume all the irrational values lying in the $[m, M]$. But since $f(x)$ takes only rational values, we must have $m = M$ i.e., $f(x)$ must be a constant function. As $f(2) = 10$, we get

$$f(x) = 10 \quad \forall x \in [1, 3] \Rightarrow f(1.5) = 10$$

B. True/False

1. Consider $f(x) = \frac{|x-a|}{x-a}, g(x) = \frac{x-a}{|x-a|}$

then $\lim_{x \rightarrow a} (f(x)g(x))$ exists but neither $\lim_{x \rightarrow a} f(x)$ nor

$\lim_{x \rightarrow a} g(x)$ exists.

C. MCQs with ONE Correct Answer

1. (c) $f(x) = \sqrt{\frac{x - \sin x}{x + \cos^2 x}}$

$$\Rightarrow \lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \sqrt{\frac{1 - \frac{\sin x}{x}}{1 + \frac{\cos^2 x}{x}}} = \sqrt{\frac{1-0}{1+0}} = 1$$

2. (d) $f(x) = \frac{\tan(\pi[x-\pi])}{1+[x]^2}$

By def. $[x-\pi]$ is an integer whatever be the value of x . And so $\pi[x-\pi]$ is an integral multiple of π .

Consequently $\tan(\pi[x-\pi]) = 0, \forall x$.

And since $1 + [x]^2 \neq 0$ for any x , we conclude that $f(x) = 0$.

Limits, Continuity and Differentiability

Thus $f(x)$ is constant function and so, it is continuous and differentiable any no. of times, that is $f'(x), f''(x), f'''(x), \dots$ all exist for every x , their value being 0 at every pt. x . Hence, out of all the alternatives only (d) is correct.

3. (a) $f(x) = e^{-x}$ is one such function.

Here $f(0) = 1, f'(0) = -1, f(x) > 0, \forall x$.

$$\therefore f''(x) > 0 \quad \forall x$$

4. (d)
$$\lim_{x \rightarrow 1} \frac{-\sqrt{25-x^2} - (-\sqrt{24})}{x-1}$$

$$= \lim_{x \rightarrow 1} \frac{\sqrt{24} - \sqrt{25-x^2}}{x-1} \times \frac{\sqrt{24} + \sqrt{25-x^2}}{\sqrt{24} + \sqrt{25-x^2}}$$

$$= \lim_{x \rightarrow 1} \frac{x^2 - 1}{(x-1)[\sqrt{24} + \sqrt{25-x^2}]}$$

$$= \lim_{x \rightarrow 1} \frac{x+1}{[\sqrt{24} + \sqrt{25-x^2}]} = \frac{2}{2\sqrt{24}} = \frac{1}{2\sqrt{6}}$$

5. (c)
$$\lim_{x \rightarrow a} \frac{g(x)f(a) - g(a)f(x)}{x-a}$$

$$= \lim_{h \rightarrow 0} \frac{g(a+h)f(a) - g(a)f(a+h)}{h} \quad [\text{For } x = a+h]$$

$$= \lim_{h \rightarrow 0} \frac{g(a+h)f(a) - g(a)f(a) + g(a)f(a) - g(a)f(a+h)}{h}$$

$$= \lim_{h \rightarrow 0} f(a) \left[\frac{g(a+h) - g(a)}{h} \right] - \lim_{h \rightarrow 0} g(a) \left[\frac{f(a+h) - f(a)}{h} \right]$$

$$= f(a)g'(a) - g(a)f'(a) = 2 \times 2 - (-1) \times 1 = 5$$

6. (b) For $f(x)$ to be continuous at $x=0$

$$f(0) = \lim_{x \rightarrow 0} f(x)$$

$$= \lim_{x \rightarrow 0} \frac{\ln(1+ax) - \ln(1-bx)}{x}$$

$$\left[\text{Using } \lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1 \right]$$

$$= a + b$$

7. (b)
$$\lim_{n \rightarrow \infty} \left(\frac{1}{1-n^2} + \frac{2}{1-n^2} + \dots + \frac{n}{1-n^2} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1+2+3+\dots+n}{1-n^2} = \lim_{n \rightarrow \infty} \frac{\Sigma n}{1-n^2} = \lim_{n \rightarrow \infty} \frac{n(n+1)}{2(1-n^2)}$$

$$= \lim_{n \rightarrow \infty} \frac{1+1/n}{2 \left[\frac{1}{n^2} - 1 \right]} = -1/2$$

8. (d) The given function can be restated as

$$f(x) = \begin{cases} \frac{\sin[x]}{[x]}, & \text{if } x \in (-\infty, 0) \cup [1, \infty) \\ 0, & \text{if } x \in [0, 1) \end{cases}$$

$$\therefore \lim_{x \rightarrow 0^-} f(x) = \lim_{h \rightarrow 0} \frac{\sin[-h]}{[-h]}$$

$$= \lim_{x \rightarrow 0^-} \frac{\sin(-1)}{(-1)} = \sin 1$$

$$\text{And } \lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0} 0 = 0$$

$$\therefore \lim_{x \rightarrow 0^-} f(x) \neq \lim_{x \rightarrow 0^+} f(x) \text{ as } \sin 1 \neq 0$$

$$\therefore \lim_{x \rightarrow 0} f(x) \text{ does not exist.}$$

9. (a) We have $f: R \rightarrow R$, a differentiable function and $f(1) = 4$

NOTE THIS STEP

$$\lim_{x \rightarrow 1} \int_4^{f(x)} \frac{2t}{x-1} dt = \lim_{x \rightarrow 1} \left[\frac{t^2}{x-1} \right]_4^{f(x)}$$

$$= \lim_{x \rightarrow 1} \frac{(f(x))^2 - 16}{x-1} = \lim_{x \rightarrow 1} \frac{f(x)-4}{x-1} \cdot \lim_{x \rightarrow 1} (f(x)+4)$$

$$= f'(1) \cdot (f(1)+4) = 8f'(1) \quad [\text{Using } f(1)=4]$$

10. (b) We have $f(x) = [\tan^2 x]$

$\tan x$ is an increasing function for $-\frac{\pi}{4} < x < \frac{\pi}{4}$

$$\therefore \tan\left(-\frac{\pi}{4}\right) < \tan x < \tan\left(\frac{\pi}{4}\right) \quad \text{NOTE THIS STEP}$$

$$\Rightarrow -1 < \tan x < 1 \Rightarrow 0 < \tan^2 x < 1$$

$$\Rightarrow [\tan^2 x] = 0$$

$$\text{Hence, } \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} [\tan^2 x] = 0$$

Also $f(0) = 0$

$\therefore f(x)$ is continuous at $x=0$

11. (c) When x is not an integer, both the functions $[x]$ and

$$\cos\left(\frac{2x-1}{2}\right)\pi \text{ are continuous.}$$

$\therefore f(x)$ is continuous on all non integral points.

For $x = n \in I$

$$\lim_{x \rightarrow n^-} f(x) = \lim_{x \rightarrow n^-} [x] \cos\left(\frac{2x-1}{2}\right)\pi$$

$$= (n-1) \cos\left(\frac{2n-1}{2}\right)\pi = 0$$

$$\lim_{x \rightarrow n^+} f(x) = \lim_{x \rightarrow n^+} [x] \cos\left(\frac{2x-1}{2}\right)\pi$$

$$= n \cos\left(\frac{2n-1}{2}\right)\pi = 0$$

$$\text{Also } f(n) = n \cos\left(\frac{2n-1}{2}\right)\pi = 0$$

$\therefore f$ is continuous at all integral pts as well.

Thus, f is continuous everywhere.

12. (b) We have $= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{2n} \frac{r}{\sqrt{n^2 + r^2}}$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{2n} \frac{r}{n\sqrt{1 + (r/n)^2}}$$

$$= \int_0^2 \frac{x}{\sqrt{1+x^2}} dx \left[\because \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{a_n} f\left(\frac{r}{n}\right) = \int_0^a f(x) dx \right]$$

$$= \left[\sqrt{1+x^2} \right]_0^2 = \sqrt{5} - 1$$

13. (d) We have
- $f(x) = [x]^2 - [x^2]$

At $x = 0$,

$$\begin{aligned} \text{L.H.L.} &= \lim_{h \rightarrow 0} f(-h) = \lim_{h \rightarrow 0} [-h]^2 - [(-h)^2] \\ &= \lim_{h \rightarrow 0} f(-1)^2 - [h^2] = \lim_{h \rightarrow 0} 1 - 0 = 1 \end{aligned}$$

$$\begin{aligned} \text{R.H.L.} &= \lim_{h \rightarrow 0} f(h) = \lim_{h \rightarrow 0} [h]^2 - [h^2] \\ &= \lim_{h \rightarrow 0} 0 - 0 = 0 \end{aligned}$$

 $\therefore$ L.H.L. $\neq$ R.H.L. $\therefore f(x)$ is not continuous at $x = 0$.At $x = 1$

$$\begin{aligned} \text{L.H.L.} &= \lim_{h \rightarrow 0} f(1-h) = \lim_{h \rightarrow 0} [1-h]^2 - [(1-h)^2] \\ &= \lim_{h \rightarrow 0} 0 - 0 = 0 \end{aligned}$$

$$\begin{aligned} \text{R.H.L.} &= \lim_{h \rightarrow 0} f(1+h) = \lim_{h \rightarrow 0} [1+h]^2 - [(1+h)^2] \\ &= \lim_{h \rightarrow 0} 1 - 1 = 0 \end{aligned}$$

$$f(1) = [1]^2 - [1^2] = 1 - 1 = 0$$

 $\therefore$ L.H.L. = R.H.L. = $f(1)$ $\therefore f(x)$ is continuous at $x = 1$.Clearly at other integral pts $f(x)$ is not continuous.

14. (d) We have
- $|x| = \begin{cases} -x & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases}$

$$\text{Also, } |x^2 - 3x + 2| = |(x-1)(x-2)|$$

$$= \begin{cases} (1-x)(2-x) & \text{if } x < 1 \\ (x-1)(2-x) & \text{if } 1 \leq x < 2 \\ (x-1)(x-2) & \text{if } x \geq 2 \end{cases}$$

As $\cos(-\theta) = \cos \theta \Rightarrow \cos |x| = \cos x$ $\therefore$ Given function can be written as

$$f(x) = \begin{cases} (x^2 - 1)(x-1)(x-2) + \cos x & \text{if } x \leq 1 \\ -(x^2 - 1)(x-1)(x-2) + \cos x & \text{if } 1 \leq x < 2 \\ (x^2 - 1)(x-1)(x-2) + \cos x & \text{if } x \geq 2 \end{cases}$$

This function is differentiable at all points except possibly at $x = 1$ and $x = 2$.

$$\begin{aligned} \text{Lf}'(1) &= \left\{ \frac{d}{dx} [(x^2 - 1)(x-1)(x-2) + \cos x] \right\}_{x=1} \\ &= -\sin 1 \end{aligned}$$

$$\begin{aligned} \text{Rf}'(1) &= \left\{ \frac{d}{dx} [-(x^2 - 1)(x-1)(x-2) + \cos x] \right\}_{x=1} \\ &= -\sin 1 \end{aligned}$$

 $\therefore$ Lf'(1) = Rf'(1) $\therefore f$ is differentiable at $x = 1$.

$$\begin{aligned} \text{Lf}'(2) &= \left\{ \frac{d}{dx} [-(x^2 - 1)(x-1)(x-2) + \cos x] \right\}_{x=2} \\ &= -3 - \sin 2 \end{aligned}$$

$$\begin{aligned} \text{Rf}'(2) &= \left\{ \frac{d}{dx} [(x^2 - 1)(x-1)(x-2) + \cos x] \right\}_{x=2} \\ &= 3 - \sin 2 \end{aligned}$$

$$\text{Lf}'(2) \neq \text{Rf}'(2)$$

 $\therefore f$ is not differentiable at $x = 2$.

$$\begin{aligned} 15. (c) \quad \lim_{x \rightarrow 0} \frac{x \tan 2x - 2x \tan x}{(1 - \cos 2x)^2} \\ = \lim_{x \rightarrow 0} \frac{x \left\{ 2x + \frac{8x^3}{3} + \frac{64x^5}{15} + \dots \right\} - 2x \left\{ x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots \right\}}{4 \sin^4 x} \\ = \lim_{x \rightarrow 0} \frac{x^4 \left\{ \frac{8}{3} - \frac{2}{3} + \text{terms containing higher positive powers of } x \right\}}{4 \sin^4 x} \\ = \frac{1}{4} \cdot 2 = \frac{1}{2} \end{aligned}$$

16. (c) For
- $x \in R$
- ,

$$\begin{aligned} \lim_{x \rightarrow \infty} \left(\frac{x-3}{x+2} \right)^x &= \lim_{x \rightarrow \infty} \left\{ \left[1 - \frac{5}{x+2} \right]^{\frac{-(x+2)}{5}} \right\}^{\frac{-5x}{x+2}} \\ &= e^{\lim_{x \rightarrow \infty} -\frac{5}{1 + \frac{2}{x}}} = e^{-5} \end{aligned}$$

$$17. (b) \quad \lim_{x \rightarrow 0} \frac{\sin(\pi \cos^2 x)}{x^2} = \lim_{x \rightarrow 0} \frac{\sin(\pi - \pi \sin^2 x)}{x^2}$$

$$[\sin(\pi - \theta) = \sin \theta]$$

$$= \lim_{x \rightarrow 0} \frac{\sin(\pi \sin^2 x)}{\pi \sin^2 x} \times \frac{(\pi \sin^2 x)}{x^2} = \pi$$

18. (a) At
- $LHD = \lim_{x \rightarrow k} \frac{f(k) - f(k-h)}{h}$
- (
- $k = \text{integer}$
-)

$$= \lim_{h \rightarrow 0} \frac{[k] \sin k\pi - [k-h] \sin(k-h)\pi}{h}$$

$$= \lim_{h \rightarrow 0} \frac{-(k-1) \sin(k-h)\pi}{h} \quad [\because \sin k\pi = 0]$$

$$= \lim_{h \rightarrow 0} \frac{-(k-1) \sin(k\pi - h\pi)}{h}$$

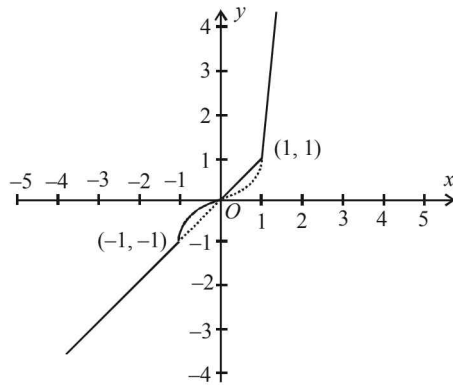
$$[\sin(k\pi - \theta) = (-1)^{k-1} \sin \theta]$$

$$= \lim_{h \rightarrow 0} \frac{-(k-1)(-1)^{k-1} \sin h\pi}{h\pi} \times \pi = \pi(k-1)(-1)^k$$

19. (d)
- $f(x) = \max. \{x, x^3\}$

$$= \begin{cases} x; & x < -1 \\ x^3; & -1 \leq x \leq 0 \\ x; & 0 \leq x \leq 1 \\ x^3; & x \geq 1 \end{cases}$$

KEY CONCEPTA continuous function $f(x)$ is not differentiable at $x = a$ if graphically it takes a sharp turn at $x = a$.Graph of $f(x) = \max \{x, x^3\}$ is as shown with solid lines.



From graph of $f(x)$ at $x = -1, 0, 1$, we have sharp turns.

$\therefore f(x)$ is not differentiable at $x = -1, 0, 1$.

20. (d) Let us test each of four options.

$$(a) f(x) = \cos |x| + |x| = \begin{cases} \cos x - x, & x < 0 \\ \cos x + x, & x \geq 0 \end{cases}$$

$$f'(x) = \begin{cases} -\sin x - 1, & x < 0 \\ -\sin x + 1, & x \geq 0 \end{cases}$$

At $x = 0$, $LD = -1$, $RD = 1$

$\therefore$ Not differentiable

$$(b) f(x) = \cos |x| - |x| = \begin{cases} \cos x + x, & x < 0 \\ \cos x - x, & x \geq 0 \end{cases}$$

$\therefore$ Not differentiable at $x = 0$

$$(c) f(x) = \sin |x| + |x| = \begin{cases} -\sin x - x, & x < 0 \\ \sin x - x, & x \geq 0 \end{cases}$$

$\therefore$ Not differentiable at $x = 0$

$$(d) f(x) = \sin |x| - |x| = \begin{cases} -\sin x + x, & x < 0 \\ \sin x - x, & x \geq 0 \end{cases}$$

$$f'(x) = \begin{cases} -\cos x + 1, & x < 0 \\ \cos x - 1, & x \geq 0 \end{cases}$$

At $x = 0$, $LD = 0$, $RD = 0$

$\therefore f$ is differentiable at $x = 0$.

21. (d) The given function is

$$f(x) = \begin{cases} \tan^{-1} x & \text{if } |x| \leq 1 \\ \frac{1}{2}(|x| - 1) & \text{if } |x| > 1 \end{cases}$$

$$\Rightarrow f(x) = \begin{cases} \frac{1}{2}(-x - 1) & \text{if } x < -1 \\ \tan^{-1} x & \text{if } -1 \leq x \leq 1 \\ \frac{1}{2}(x - 1) & \text{if } x > 1 \end{cases}$$

Clearly L.H.L. at $(x = -1) = \lim_{h \rightarrow 0} f(-1 - h) = 0$

R.H.L. at $(x = -1) = \lim_{h \rightarrow 0} f(-1 + h)$

$$= \lim_{h \rightarrow 0} \tan^{-1}(-1 + h) = 3\pi/4$$

$\therefore$ L.H.L. $\neq$ R.H.L. at $x = -1$

$\therefore f(x)$ is discontinuous at $x = -1$

Also we can prove in the same way, that $f(x)$ is discontinuous at $x = 1$

$\therefore f'(x)$ can not be found for $x = \pm 1$ or domain of

$$f'(x) = R - \{-1, 1\}$$

22. (c) Given that,

$$\lim_{x \rightarrow 0} \frac{(\cos x - 1)(\cos x - e^x)}{x^n} = \text{finite non zero number}$$

$$= \lim_{x \rightarrow 0} \frac{(1 - \cos x)(1 + \cos x)(e^x - \cos x)}{x^n(1 + \cos x)}$$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin^2 x}{x^2} \right) \cdot \left(\frac{e^x - \cos x}{x^{n-2}} \right) \cdot \left(\frac{1}{1 + \cos x} \right)$$

$$= \lim_{x \rightarrow 0} 1^2 \cdot \frac{e^x - \cos x}{x^{n-2}} \cdot \frac{1}{2}$$

$$= \frac{1}{2} \lim_{x \rightarrow 0} \frac{e^x - \sin x}{(x - 2)x^{n-3}} \quad [\text{using L' Hospital's rule}]$$

For this limit to be finite, $n - 3 = 0 \Rightarrow n = 3$

23. (c) Given that $f: R \rightarrow R$ such that

$f(1) = 3$ and $f'(1) = 6$

$$\text{Then } \lim_{x \rightarrow 0} \left[\frac{f(1+x)}{f(1)} \right]^{1/x}$$

$$= e^{\lim_{x \rightarrow 0} \frac{1}{x} [\log f(1+x) - \log f(1)]}$$

$$= e^{\lim_{x \rightarrow 0} \frac{\frac{1}{f(1+x)} f'(1+x)}{1}} \quad [\text{Using L' Hospital rule}]$$

$$= e^{\frac{f'(1)}{f(1)}} = e^{6/3} = e^2$$

24. (d) We are given that

$$\lim_{x \rightarrow 0} \frac{[(a-n)nx - \tan x] \sin nx}{x^2} = 0$$

where n is non zero real number

$$\Rightarrow \lim_{x \rightarrow 0} n \cdot \frac{\sin nx}{nx} \left[\left\{ (a-n)n - \frac{\tan x}{x} \right\} \right] = 0$$

$$\Rightarrow 1 \cdot n [(a-n)n - 1] = 0 \Rightarrow a = \frac{1}{n} + n$$

25. (d) Let $L = \lim_{h \rightarrow 0} \frac{f(2h+2+h^2) - f(2)}{f(h-h^2+1) - f(1)} \quad \left[\frac{0}{0} \text{ form} \right]$

$\therefore$ Applying L Hospital's rule, we get

$$L = \lim_{h \rightarrow 0} \frac{f'(2h+2+h^2) \cdot (2+2h)}{f'(h-h^2+1) \cdot (1-2h)}$$

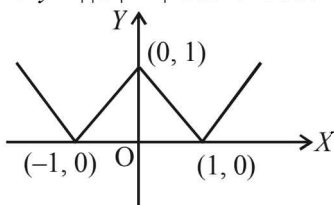
$$= \frac{f'(2) \cdot 2}{f'(1) \cdot 1} = \frac{6 \times 2}{4 \times 1} = 3$$

26. (c) Let $L = \lim_{x \rightarrow 0} \frac{f(x^2) - f(x)}{f(x) - f(0)} \quad \left[\because f'(a) > 0, f \text{ being strictly increasing} \right]$

Using L.H. Rule, we get

$$L = \lim_{x \rightarrow 0} \frac{f'(x^2) \cdot 2x - f'(x)}{f'(x)} = \lim_{x \rightarrow 0} \frac{f'(x^2) \cdot 2x}{f'(x)} - 1 = 0 - 1 = -1$$

27. (a) Graph of $y = ||x| - 1|$ is as follows :



The graph has sharp turnings at $x = -1, 0$ and 1 ; and hence not differentiable at $x = -1, 0, 1$.

28. (b) Given that $f(x)$ is a continuous and differentiable function and $f\left(\frac{1}{x}\right) = 0, x = n, n \in I$

$$\therefore f(0^+) = f\left(\frac{1}{\infty}\right) = 0$$

Since R.H.L. = 0,

$\therefore f(0) = 0$ for $f(x)$ to be continuous.

$$\text{Also } f'(0) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h - 0} = \lim_{h \rightarrow 0} \frac{f(h)}{h} = 0$$

= 0 [Using $f(0) = 0$ and $f(0^+) = 0$]

Hence $f(0) = 0, f'(0) = 0$

29. (c) $\lim_{x \rightarrow 0} [(\sin x)^{1/x} + (1/x)^{\sin x}]$

$$\begin{aligned} &= \lim_{x \rightarrow 0} (\sin x)^{1/x} + \lim_{x \rightarrow 0} \left(\frac{1}{x}\right)^{\sin x} \\ &= \lim_{x \rightarrow 0} \sin x \log\left(\frac{1}{x}\right) \\ &= 0 + e^{\lim_{x \rightarrow 0} \sin x \log\left(\frac{1}{x}\right)} \\ &\quad [\because |\sin x| < 1 \text{ when } x \rightarrow 0] \\ &= e^{\lim_{x \rightarrow 0} \frac{-\log x}{\operatorname{cosec} x}} = e^{\lim_{x \rightarrow 0} \frac{-1/x}{-\operatorname{cosec} x \cot x}} \\ &\quad \text{[Using L' Hospital rule]} \end{aligned}$$

$$= e^{\lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \tan x} = e^0 = 1$$

30. (a) Given that $f(x)$ is differentiable on $(0, \infty)$ with

$$f(1) = 1 \text{ and } \lim_{t \rightarrow x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1 \text{ for each } x > 0$$

$$\Rightarrow \lim_{t \rightarrow x} \frac{2t f(x) - x^2 f'(t)}{1} = 1 \text{ [Using L' Hospital rule]}$$

$$\Rightarrow 2x f(x) - x^2 f'(x) = 1 \Rightarrow f'(x) - \frac{2}{x} f(x) = -\frac{1}{x^2}$$

[Linear differential equation]

Integrating factor

NOTE THIS STEP

$$e^{\int -\frac{2}{x} dx} = e^{-2 \log x} = e^{\log 1/x^2} = \frac{1}{x^2}$$

$$\therefore \text{Solution is } f(x) \times \frac{1}{x^2} = \int \left(-\frac{1}{x^2}\right) \times \frac{1}{x^2} dx$$

$$\Rightarrow \frac{f(x)}{x^2} = \frac{1}{3x^3} + C \Rightarrow f(x) = Cx^2 + \frac{1}{3x}$$

Also $f(1) = 1$

$$\Rightarrow 1 = C + \frac{1}{3} \Rightarrow C = 2/3 \therefore f(x) = \frac{2}{3}x^2 + \frac{1}{3x}$$

31. (a) **KEY CONCEPT**

$$\begin{aligned} \frac{d}{dx} \left[\int_{g(x)}^{h(x)} f(t) dt \right] &= f(h(x)) h'(x) - f(g(x)) g'(x) \\ &= \frac{f(2) \times 2 \times 2 \times 1}{2 \times \frac{\pi}{4}} = \frac{8}{\pi} f(2) \end{aligned}$$

$$\text{Let } L = \lim_{x \rightarrow \frac{\pi}{4}} \frac{\int_2^{\sec^2 x} f(t) dt}{x^2 - \frac{\pi^2}{16}} \quad \left[\frac{0}{0} \text{ form} \right]$$

On applying L' Hospital's rule, we get

$$\begin{aligned} L &= \lim_{x \rightarrow \frac{\pi}{4}} \frac{\frac{d}{dx} \left[\int_2^{\sec^2 x} f(t) dt \right]}{\frac{d}{dx} \left(x^2 - \frac{\pi^2}{16} \right)} \\ L &= \lim_{x \rightarrow \frac{\pi}{4}} \frac{f(\sec^2 x) \cdot 2 \sec^2 x \tan x}{2x} \end{aligned}$$

32. (c) As per question,

$p =$ left hand derivative of $|x-1|$ at $x = 1 \Rightarrow p = -1$

$$\text{Also } \lim_{x \rightarrow 1^+} g(x) = p$$

$$\text{Where } g(x) = \frac{(x-1)^n}{\log \cos^m(x-1)}, \quad 0 < x < 2,$$

m, n are integers, $m \neq 0, n > 0$

$\therefore$ we get,

$$\lim_{x \rightarrow 1^+} \frac{(x-1)^n}{\log \cos^m(x-1)} = -1 \Rightarrow \lim_{h \rightarrow 0} \frac{h^n}{\log \cos^m h} = -1$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{h^n}{m(\log \cos h)} = -1 \text{ [Using L' Hospital's rule]}$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{n h^{n-1} \cos h}{m(-\sin h)} = -1 \text{ [Using L' Hospital's rule]}$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{n h^{n-2} \cos h}{m \left(\frac{\sin h}{h} \right)} = 1 \Rightarrow n = 2 \text{ and } m = 2$$

33. (d) $\lim_{x \rightarrow 0} [1 + x \ell n(1 + b^2)]^{\frac{1}{x}} = 2b \sin^2 \theta$

$$\Rightarrow e^{\lim_{x \rightarrow 0} \frac{1}{x} \ell n[1 + x \ell n(1 + b^2)]} = 2b \sin^2 \theta$$

$$\Rightarrow e^{\lim_{x \rightarrow 0} \frac{\ell n[1 + x \ell n(1 + b^2)]}{x \ell n(1 + b^2)} \times \ell n(1 + b^2)} = 2b \sin^2 \theta$$

$$\Rightarrow e^{\ell n(1 + b^2)} = 2b \sin^2 \theta$$

$$\Rightarrow 1 + b^2 = 2b \sin^2 \theta \Rightarrow 2 \sin^2 \theta = b + \frac{1}{b}$$

We know that $2 \sin^2 \theta \leq 2$ and $b + \frac{1}{b} \geq 2$ for $b > 0$

$$\therefore 2 \sin^2 \theta = b + \frac{1}{b} = 2 \Rightarrow \sin^2 \theta = 1$$

$$\text{As } \theta \in (-\pi, \pi], \therefore \theta = \pm \frac{\pi}{2}$$

$$\begin{aligned} 34. \quad (b) \quad \text{Given: } \lim_{x \rightarrow \infty} \left(\frac{x^2 + x + 1}{x + 1} - ax - b \right) &= 4 \\ \Rightarrow \lim_{x \rightarrow \infty} \frac{x^2 + x + 1 - ax^2 - ax - bx - b}{x + 1} &= 4 \\ \Rightarrow \lim_{x \rightarrow \infty} \frac{(1-a)x^2 + (1-a-b)x + (1-b)}{x + 1} &= 4 \end{aligned}$$

For this limit to be finite $1-a=0 \Rightarrow a=1$
then given limit reduces to

$$\lim_{x \rightarrow \infty} \frac{-bx + (1-b)}{x + 1} = 4 \Rightarrow \lim_{x \rightarrow \infty} \frac{-b + \frac{(1-b)}{x}}{1 + \frac{1}{x}} = 4$$

$$\Rightarrow -b = 4 \text{ or } b = -4$$

Hence $a = 1, b = -4$.

$$\begin{aligned} 35. \quad (b) \quad \text{We have } f'(0^+) &= \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{h^2 \left| \cos \frac{\pi}{h} \right|}{h} = \lim_{h \rightarrow 0} h \left| \cos \frac{\pi}{h} \right| \\ &= 0 \times \text{some finite value} = 0 \end{aligned}$$

$$\begin{aligned} \text{Also, } f'(0^-) &= \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} = \lim_{h \rightarrow 0} \frac{h^2 \left| \cos \frac{\pi}{-h} \right|}{-h} \\ &= \lim_{h \rightarrow 0} -h \left| \cos \frac{\pi}{h} \right| = 0 \times \text{some finite value} = 0 \end{aligned}$$

$\therefore f'(0^+) = f'(0^-) \Rightarrow f$ is differentiable at $x = 0$

$$\begin{aligned} \text{Now } f'(2^+) &= \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(2+h)^2 \left| \cos \frac{\pi}{2+h} \right| - 4 \left| \cos \frac{\pi}{2} \right|}{h} \\ &= \lim_{h \rightarrow 0} \frac{(2+h)^2 \left(\cos \frac{\pi}{2+h} \right)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(2+h)^2}{h} \sin \left(\frac{\pi}{2} - \frac{\pi}{2+h} \right) \\ &= \lim_{h \rightarrow 0} \frac{(2+h)^2}{h} \sin \left(\frac{\pi h}{2(2+h)} \right) \\ &= \lim_{h \rightarrow 0} \frac{(2+h)^2}{h} \times \frac{\sin \left(\frac{\pi h}{2(2+h)} \right)}{\left(\frac{\pi h}{2(2+h)} \right)} \times \frac{\pi h}{2(2+h)} = \pi \end{aligned}$$

$$\begin{aligned} \text{Also } f'(2^-) &= \lim_{h \rightarrow 0} \frac{f(2-h) - f(2)}{-h} \\ &= \lim_{h \rightarrow 0} \frac{(2-h)^2 \left| \cos \left(\frac{\pi}{2-h} \right) \right| - 0}{-h} \\ &= \lim_{h \rightarrow 0} \frac{-(2-h)^2 \cos \left(\frac{\pi}{2-h} \right)}{-h} \\ &= \lim_{h \rightarrow 0} \frac{(2-h)^2 \sin \left(\frac{\pi}{2} - \frac{\pi}{2-h} \right)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(2-h)^2}{h} \times \frac{\sin \left(\frac{-\pi h}{2(2-h)} \right)}{\left(\frac{-\pi h}{2(2-h)} \right)} \times \left(\frac{-\pi h}{2(2-h)} \right) = -\pi \end{aligned}$$

As $f'(2^+) \neq f'(2^-) \Rightarrow f$ is not differentiable at $x = 2$.

36. (b) The given equation is

$$(\sqrt[3]{1+a} - 1)x^2 + (\sqrt{1+a} - 1)x + (\sqrt[6]{1+a} - 1) = 0$$

Let $a + 1 = y$, then equation reduces to
 $(y^{1/3} - 1)x^2 + (y^{1/2} - 1)x + (y^{1/6} - 1) = 0$

Dividing both sides by $y - 1$, we get

$$\left(\frac{y^{1/3} - 1}{y - 1} \right) x^2 + \left(\frac{y^{1/2} - 1}{y - 1} \right) x + \left(\frac{y^{1/6} - 1}{y - 1} \right) = 0$$

Taking limit as $y \rightarrow 1$ i.e. $a \rightarrow 0$ on both sides we get

$$\frac{1}{3}x^2 + \frac{1}{2}x + \frac{1}{6} = 0 \Rightarrow 2x^2 + 3x + 1 = 0$$

$$\Rightarrow x = -1, -\frac{1}{2} \text{ (roots of the equation)}$$

$$\text{Thus } \lim_{a \rightarrow 0^+} \alpha(a) = -1, \lim_{a \rightarrow 0^+} \beta(a) = -\frac{1}{2}$$

D. MCQs with ONE or MORE THAN ONE Correct

1. (a, b, d) Given that $x + |y| = 2y$

If $y < 0$ then $x - y = 2y$

$$\Rightarrow y = x/3 \Rightarrow x < 0$$

If $y = 0$ then $x = 0$. If $y > 0$ then $x + y = 2y$

$$\Rightarrow y = x \Rightarrow x > 0$$

Thus we can define $f(x) = y = \begin{cases} x/3, & x < 0 \\ x, & x \geq 0 \end{cases}$

Continuity at $x = 0$

$$LL = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} (-h/3) = 0$$

$$RL = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} h = 0$$

$$f(0) = 0$$

$$\text{As } LL = RL = f(0)$$

$\therefore f(x)$ is continuous at $x = 0$

Differentiability at $x = 0$

$$Lf' = 1/3; Rf' = 1$$

As $Lf' \neq Rf' \Rightarrow f(x)$ is not differentiable at $x=0$

But for $x < 0$, $\frac{dy}{dx} = \frac{1}{3}$.

2. (b) We have $f(x) = x(\sqrt{x} - \sqrt{x+1})$

Let us check differentiability of $f(x)$ at $x=0$

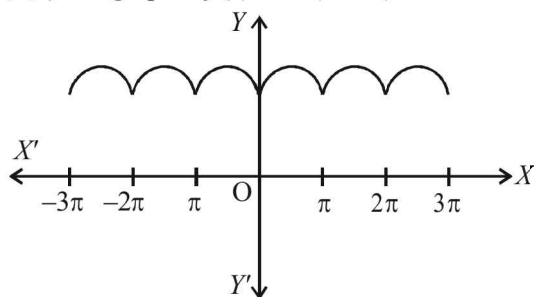
$$\begin{aligned} Lf'(0) &= \lim_{h \rightarrow 0} \frac{(0-h)[\sqrt{0-h} - \sqrt{0-h+1}] - 0}{-h} \\ &= \lim_{h \rightarrow 0} [\sqrt{-h} - \sqrt{-h+1}] = 0 - \sqrt{1} = -1 \end{aligned}$$

$$\begin{aligned} Rf'(0) &= \lim_{h \rightarrow 0} \frac{(0+h)[\sqrt{0+h} - \sqrt{0+h+1}] - 0}{h} \\ &= \lim_{h \rightarrow 0} \sqrt{h} - \sqrt{h+1} = -1 \end{aligned}$$

Since $Lf'(0) = Rf'(0)$

$\therefore f$ is differentiable at $x=0$.

3. (b,d,e) The graph of $f(x) = 1 + |\sin x|$ is as shown in the fig.



From graph it is clear that function is continuous every where but not differentiable at integral multiples of π ($\because$ at these points curve has sharp turnings)

4. (a,b,d) We have, for $-1 \leq x \leq 1 \Rightarrow 0 \leq x \sin \pi x \leq 1/2$

$\therefore f(x) = [x \sin \pi x] = 0$

Also $x \sin \pi x$ becomes negative and numerically less than 1 when x is slightly greater than 1 and so by definition of $[x]$,

$$f(x) = [x \sin \pi x] = -1 \text{ when } 1 < x < 1 + h$$

Thus $f(x)$ is constant and equal to 0 in the closed interval $[-1, 1]$ and so $f(x)$ is continuous and differentiable in the open interval $(-1, 1)$.

At $x=1$, $f(x)$ is clearly discontinuous, since $f(1-0) = 0$ and $f(1+0) = -1$ and $f(x)$ is non-differentiable at $x=1$.

5. (a) The given function is,

$$f(x) = \frac{x}{1+|x|} = \begin{cases} \frac{x}{1-x}, & x < 0 \\ \frac{x}{1+x}, & x \geq 0 \end{cases}$$

$$\text{For } x < 0, f'(x) = \frac{1(1-x) - (-1)x}{(1-x)^2} = \frac{1}{(1-x)^2},$$

$$\text{For } x > 0, f'(x) = \frac{1(1+x) - 1(x)}{(1+x)^2} = \frac{1}{(1+x)^2},$$

For $x=0$,

$$Lf'(0) = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} = \lim_{h \rightarrow 0} \frac{\frac{-h}{1+h} - 0}{-h} = 1$$

$$Rf'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{\frac{h}{1+h} - 0}{h} = 1$$

$$Lf'(0) = Rf'(0)$$

$\Rightarrow f$ is differentiable at $x=0$

Hence f is differentiable in $(-\infty, \infty)$.

$$6. \text{ (a,b,c) } f(x) = \begin{cases} |x-3|, & x \geq 1 \\ \frac{x^2}{4} - \frac{3x}{2} + \frac{13}{4}, & x < 1 \end{cases}$$

$$= \begin{cases} \frac{x^2}{4} - \frac{3x}{2} + \frac{13}{4}, & x < 1 \\ 3-x, & 1 \leq x < 3 \\ x-3, & x \geq 3 \end{cases}$$

$$Lf'(1) = \frac{2x}{4} - \frac{3}{2} = -1$$

$$Rf'(1) = -1. \text{ Thus } Lf'(1) = Rf'(1)$$

$\therefore f$ is differentiable at $x=1$ and hence continuous at $x=1$.

$$\text{Again, } Lf'(3) = -1 \text{ and } Rf'(3) = 1$$

$$\Rightarrow Lf'(3) \neq Rf'(3)$$

$\Rightarrow f$ is not differentiable at $x=3$

Let us now check the continuity at $x=3$

$$\text{L.H.L.} = \lim_{h \rightarrow 0} f(3-h) = \lim_{h \rightarrow 0} [3 - (3-h)] = 0$$

$$\text{R.H.L.} = \lim_{h \rightarrow 0} f(3+h) = \lim_{h \rightarrow 0} [3 + h - 3] = 0$$

$$f(3) = 0$$

$\therefore f$ is continuous at $x=3$

7. (b) We have, $f(x) = \frac{x}{2} - 1$

$$\therefore [f(x)] = \left[\frac{x}{2} - 1 \right] = -1, 0 \leq x < 2$$

$$\tan [f(x)] = \tan (-1), 0 \leq x < 2 = 0, 2 \leq x \leq \pi$$

$\therefore$ The function $\tan [f(x)]$ is discontinuous at $x=2$.

$$\text{Also the function } \frac{1}{f(x)} = \frac{1}{\frac{x}{2} - 1} = \frac{2}{x-2} \text{ is}$$

discontinuous at $x=2$.

Thus both the given functions $\tan [f(x)]$ as well as

$\frac{1}{f(x)}$ are discontinuous on the interval $[0, \pi]$.

Also $f^{-1}(x) = y$

$$\Rightarrow x = f(y) = \frac{y}{2} - 1 \Leftrightarrow y = 2(x+1)$$

$\therefore f^{-1}(x) = 2(x+1)$ is continuous on $[0, \pi]$

$$8. \text{ (d) } \lim_{x \rightarrow 0} \frac{\sqrt{\frac{1}{2}(1 - \cos 2x)}}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\sqrt{\frac{1}{2} \cdot 2 \sin^2 x}}{x} = \lim_{x \rightarrow 0} \frac{|\sin x|}{x}$$

$$\begin{aligned}\therefore \text{L.H.L.} &= \lim_{h \rightarrow 0} \frac{|\sin(0-h)|}{0-h} = \lim_{h \rightarrow 0} \frac{-\sin h}{-h} \\ &= \lim_{h \rightarrow 0} \frac{\sin h}{h} = 1\end{aligned}$$

$$\text{R.H.L.} = \lim_{h \rightarrow 0} \frac{|\sin(0+h)|}{0+h} = \lim_{h \rightarrow 0} \frac{\sin h}{h} = 1$$

As L.H.L. $\neq$ R.H.L.

$\therefore$ The given limit does not exist.

9. (b,c) On $(0, \pi)$

(a) $\tan x = f(x)$

We know that $\tan x$ is discontinuous at $x = \pi/2$

$$(b) f(x) = \int_0^x t \sin\left(\frac{1}{t}\right) dt$$

NOTE THIS STEP

$$\Rightarrow f'(x) = x \sin\left(\frac{1}{x}\right) \text{ which exists on } (0, \pi)$$

$\therefore f(x)$, being differentiable, is continuous on $(0, \pi)$.

$$(c) f(x) = \begin{cases} 1 & , 0 < x \leq 3\pi/4 \\ 2 \sin \frac{2x}{9} & , 3\pi/4 < x < \pi \end{cases}$$

Clearly $f(x)$ is continuous on $(0, \pi)$ except possibly at $x = \frac{3\pi}{4}$, where,

$$\text{L.H.L.} = \lim_{h \rightarrow 0} f\left(\frac{3\pi}{4} - h\right) = \lim_{x \rightarrow 0} 1 = 1$$

$$\begin{aligned}\text{R.H.L.} &= \lim_{h \rightarrow 0} f\left(\frac{3\pi}{4} + h\right) = \lim_{x \rightarrow 0} 2 \sin \frac{2}{9} \left(\frac{3\pi}{4} + h\right) \\ &= \lim_{h \rightarrow 0} 2 \sin\left(\frac{\pi}{6} + \frac{2h}{9}\right) = 2 \sin \frac{\pi}{6} = 2 \cdot \frac{1}{2} = 1\end{aligned}$$

$$\text{Also } f\left(\frac{3\pi}{4}\right) = 1$$

$$\text{As L.H.L.} = \text{R.H.L.} = f\left(\frac{3\pi}{4}\right)$$

$\therefore f(x)$ is continuous on $(0, \pi)$

$$(d) f(x) = \begin{cases} x \sin x & , 0 < x \leq \pi/2 \\ \frac{\pi}{2} \sin(\pi + x) & , \frac{\pi}{2} < x < \pi \end{cases}$$

Here $f(x)$ will be continuous on $(0, \pi)$ if it is continuous at $x = \pi/2$.

At $x = \pi/2$,

$$\text{L.H.L.} = \lim_{h \rightarrow 0} f\left(\frac{\pi}{2} - h\right)$$

$$= \lim_{h \rightarrow 0} \left(\frac{\pi}{2} - h\right) \sin\left(\frac{\pi}{2} - h\right) = \frac{\pi}{2} \sin \frac{\pi}{2} = \frac{\pi}{2}$$

$$\text{R.H.L.} = \lim_{h \rightarrow 0} f\left(\frac{\pi}{2} + h\right) = \lim_{h \rightarrow 0} \frac{\pi}{2} \sin\left(\pi + \frac{\pi}{2} + h\right)$$

$$= \frac{\pi}{2} \sin\left(\pi + \frac{\pi}{2}\right) = \frac{-\pi}{2} \sin \frac{\pi}{2} = -\frac{\pi}{2}$$

As L.H.L. $\neq$ R.H.L.

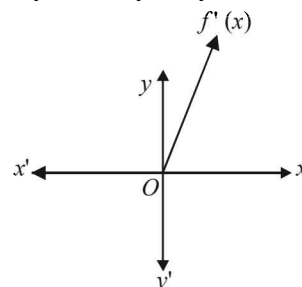
$\therefore f(x)$ is not continuous on $(0, \pi)$.

$$10. (b,c,d) f(x) = \begin{cases} 0, & x < 0 \\ x^2, & x \geq 0 \end{cases}$$

$$\therefore f'(x) = \begin{cases} 0, & x < 0 \\ 2x, & x \geq 0 \end{cases}$$

which exists $\forall x$ except possibly at $x = 0$.

At $x = 0$, $Lf' = 0 = Rf' \Rightarrow f$ is differentiable.



From graph of $f'(x)$, it is clear that $f'(x)$ is continuous but not differentiable at $x = 0$.

$$11. (a, b) \text{ We have } g(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

$$\begin{aligned}\text{If } x \neq 0, g'(x) &= x^2 \cos(1/x) \left(-\frac{1}{x^2}\right) + 2x \sin \frac{1}{x} \\ &= -\cos\left(\frac{1}{x}\right) + 2x \sin\left(\frac{1}{x}\right)\end{aligned}$$

which exists for $\forall x \neq 0$.

If $x = 0$,

$$g'(0) = \lim_{x \rightarrow 0} \frac{g(x) - g(0)}{x - 0}$$

$$= \lim_{x \rightarrow 0} \frac{x^2 \sin(1/x) - 0}{x - 0} = \lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) = 0$$

$$\Rightarrow g'(x) = \begin{cases} -\cos\left(\frac{1}{x}\right) + 2x \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

At $x = 0$, $\cos\left(\frac{1}{x}\right)$ is not continuous, therefore $g'(x)$ is not continuous at $x = 0$.

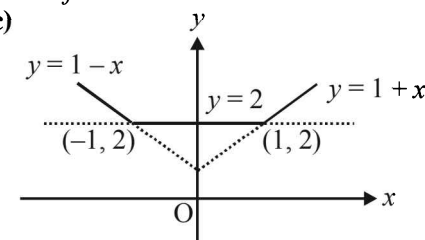
At $x = 0$

$$Lf' = \lim_{x \rightarrow 0} \frac{0 - (-x) \sin\left(-\frac{1}{x}\right)}{x} = -\sin\left(\frac{1}{x}\right)$$

which does not exist.

$\therefore f$ is not differentiable at $x = 0$.

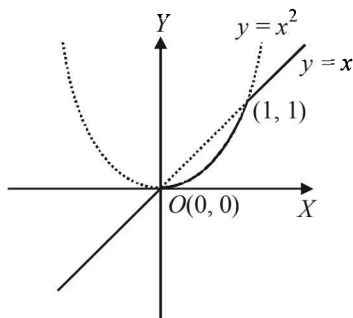
12. (a, c)



From graph it is clear that $f(x)$ is continuous every where and also differentiable everywhere except at $x = 1$ and -1 .

13. (a, c, d) From the figure it is clear that

$$h(x) = \begin{cases} x, & \text{if } x \leq 0 \\ x^2, & \text{if } 0 < x < 1 \\ x, & \text{if } x \geq 1 \end{cases}$$



From the graph it is clear that h is continuous for all $x \in \mathbb{R}$, $h'(x) = 1$ for all $x > 1$ and h is not differentiable at $x = 0$ and 1 .

14. (d) L.H.L. = $\lim_{x \rightarrow 1^-} \frac{\sqrt{1 - \cos[2(x-1)]}}{x-1}$

$$= \lim_{x \rightarrow 1^-} \frac{\sqrt{2\sin^2(x-1)}}{x-1} = \sqrt{2} \cdot \lim_{x \rightarrow 1^-} \frac{\sqrt{\sin^2(x-1)}}{x-1}$$

$$= \sqrt{2} \lim_{x \rightarrow 1^-} \frac{|\sin(x-1)|}{x-1} = \sqrt{2} \lim_{h \rightarrow 0} \frac{|\sin(-h)|}{-h}$$

$$= \sqrt{2} \lim_{h \rightarrow 0} \frac{\sin h}{-h} = -\sqrt{2}$$

Again,

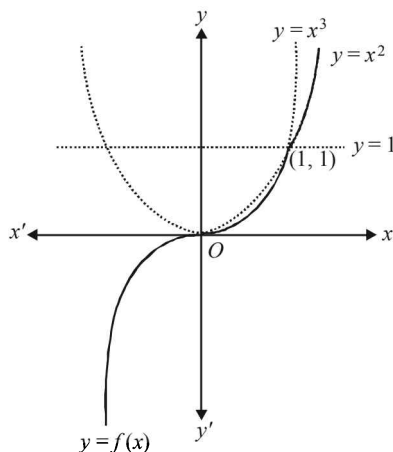
$$\text{R.H.L.} = \lim_{x \rightarrow 1^+} \frac{\sqrt{1 - \cos(x-1)}}{x-1} = \lim_{x \rightarrow 1^+} \sqrt{2} \frac{|\sin(x-1)|}{x-1}$$

Put $x = 1 + h$, $h > 0$ for $x \rightarrow 1^+$, $h \rightarrow 0$.

$$= \lim_{h \rightarrow 0} \sqrt{2} \frac{|\sin h|}{h} = \lim_{h \rightarrow 0} \sqrt{2} \frac{\sin h}{h} = \sqrt{2}$$

L.H.L. $\neq$ R.H.L. Therefore $\lim_{x \rightarrow 1} f(x)$ does not exist.

15. (a, d) From graph, $f(x)$ is continuous everywhere but not differentiable at $x = 1$.



16. (a, c) Given that $L = \lim_{x \rightarrow 0} \frac{a - \sqrt{a^2 - x^2} - \frac{x^2}{4}}{x^4}$, $a > 0$ and L is finite.

Now $L = \lim_{x \rightarrow 0} \frac{\frac{x}{\sqrt{a^2 - x^2}} - \frac{x}{2}}{4x^3}$ (Using L'Hospital's rule)

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{\sqrt{a^2 - x^2}} - \frac{1}{2}}{4x^2}$$

$\therefore L$ is finite, limiting value of numerator should be

$$\text{zero which is so when } \frac{1}{\sqrt{a^2}} - \frac{1}{2} = 0$$

$$\text{i.e. } a = 2 \quad (\because a > 0)$$

Applying L'Hospital's rule again, we get

$$L = \lim_{x \rightarrow 0} \frac{\frac{x}{(a^2 - x^2)^{3/2}}}{8x} = \lim_{x \rightarrow 0} \frac{1}{8(a^2 - x^2)^{3/2}}$$

$$= \frac{1}{8 \times a^3} = \frac{1}{8 \times 8} \quad (\text{using } a = 2)$$

$$= \frac{1}{64}$$

17. (b, c) $\because f(x+y) = f(x) + f(y) \forall x, y \in \mathbb{R}$
 $\therefore$ Putting $x = y = 0$, we get
 $f(0) = 0$

$$\text{Also } f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(h)}{h} = f'(0) = k \text{ (say)}$$

$$\Rightarrow f(x) = kx + c$$

$$\text{But } f(0) = 0 \Rightarrow c = 0$$

$$\therefore f(x) = kx$$

Which is continuous and differentiable $\forall x \in \mathbb{R}$.

$\therefore$ b and c are the correct options.

18. (a, b, c, d)

$$\text{At } x = -\frac{\pi}{2}$$

$$\text{LHL} = \lim_{x \rightarrow -\frac{\pi}{2}^-} -x - \frac{\pi}{2} = 0$$

$$\text{RHL} = \lim_{x \rightarrow -\frac{\pi}{2}^+} -\cos x = 0 \text{ and } f\left(-\frac{\pi}{2}\right) = 0$$

$$\therefore \text{LHL} = \text{RHL} = f\left(-\frac{\pi}{2}\right)$$

$$\Rightarrow f(x) \text{ is continuous at } x = -\frac{\pi}{2}$$

Also at $x = 0$

$$\text{Lf}'(0) = \sin 0 = 0; \text{Rf}'(0) = 1 - 0 = 1$$

$$\therefore \text{Lf}'(0) \neq \text{Rf}'(0)$$

$\Rightarrow f$ is not differentiable at $x = 0$

At $x = 1$

$$\text{Lf}'(1) = \text{Rf}'(1) \Rightarrow f \text{ is differentiable at } x = 1.$$

$$\text{At } x = \frac{-3}{2}, f(x) = -\cos x \text{ which is differentiable.}$$

$\therefore$ All four options are correct.

Limits, Continuity and Differentiability

19. (b,d)

$$\text{We have } f(x) = \begin{cases} a_n + \sin \pi x, & x \in [2n, 2n+1] \\ b_n + \cos \pi x, & x \in (2n-1, 2n) \end{cases}$$

As f is continuous for all n $\therefore$ At $x = 2n$, LHL = RHL = $f(2n)$

$$\Rightarrow b_n + \cos 2\pi n = a_n + \sin 2\pi n = a_n + \sin 2\pi n$$

$$\Rightarrow b_n + 1 = a_n \Rightarrow a_n - b_n = 1$$

 $\therefore b$ is correct.Also at $x = 2n+1$, LHL = RHL = $f(2n+1)$

$$\Rightarrow \lim_{h \rightarrow 0} a_n + \sin \pi(2n+1-h)$$

$$= \lim_{h \rightarrow 0} b_{n+1} + \cos \pi(2n+1-h) = a_n + \sin(2n+1)\pi$$

$$\Rightarrow a_n = b_{n+1} - 1 = a_n \Rightarrow a_n - b_{n+1} = -1$$

 $\therefore c$ is incorrect

$$\Rightarrow a_{n-1} - b_n = -1$$

 $\therefore d$ is correct.

20. (b,d)

$$\lim_{n \rightarrow \infty} \frac{1^a + 2^a + \dots + n^a}{(n+1)^{a-1}[(n+1) + (n+2) + \dots + (n+n)]} = \frac{1}{60}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{n^a}{(n+1)^{a-1}} \cdot \frac{\left(\frac{1}{n}\right)^a + \left(\frac{2}{n}\right)^a + \dots + \left(\frac{n}{n}\right)^a}{n^2 a + \frac{n(n+1)}{2}} = \frac{1}{60}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{n^{a-1}}{(n+1)^{a-1}} \cdot \frac{\frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n}\right)^a}{a + \frac{1}{2} \left(1 + \frac{1}{n}\right)} = \frac{1}{60}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \left(\frac{1}{1 + \frac{1}{n}}\right)^{a-1} \cdot \frac{\frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n}\right)^a}{a + \frac{1}{2} \left(1 + \frac{1}{n}\right)} = \frac{1}{60}$$

$$\Rightarrow \frac{\int_0^1 x^a dx}{a + \frac{1}{2}} = \frac{1}{60} \Rightarrow \frac{\left[x^{a+1}\right]_0^1}{(a+1)\left(a + \frac{1}{2}\right)} = \frac{1}{60}$$

$$\Rightarrow \frac{1}{(a+1)\left(a + \frac{1}{2}\right)} = \frac{1}{60}$$

$$\Rightarrow 2a^2 + 3a - 119 = 0 \Rightarrow a = 7 \text{ or } -\frac{17}{2}$$

21. (a,c) $g(x)$ may be discontinuous at $x = a$ or $x = b$.Let us check the continuity of $g(x)$ at $x = a$ and $x = b$

$$\lim_{x \rightarrow a^-} g(x) = 0$$

$$\lim_{x \rightarrow a^+} g(x) = \lim_{x \rightarrow a^+} \int_a^x f(t) dt = \int_a^a f(t) dt = 0$$

$$g(a) = \int_a^a f(t) dt = 0$$

 $\therefore g(x)$ is continuous at $x = a$

$$\text{Also } \lim_{x \rightarrow b^-} g(x) = \lim_{x \rightarrow b^-} \int_a^x f(t) dt = \int_a^b f(t) dt$$

$$\lim_{x \rightarrow b^+} g(x) = \int_a^b f(t) dt \Rightarrow g(b) = \int_a^b f(t) dt$$

 $\therefore g(x)$ is continuous at $x = b$ Hence $g(x)$ is continuous $\forall x \in R$

$$\text{Now } g'(x) = \begin{cases} 0, & x < a \\ f(x), & a \leq x \leq b \\ 0, & x > b \end{cases}$$

$$g'(a^-) = 0 \text{ and } g'(a^+) = f(a)$$

$$g'(b^-) = f(b) \text{ and } g'(b^+) = 0$$

As $f(a), f(b) \in [1, \infty) \therefore f(a), f(b) \neq 0$ Hence $g'(a^-) \neq g'(a^+)$ and $g'(b^-) \neq g'(b^+)$ $\therefore g$ is not differentiable at a and b .22. (a,d) Let f and g be maximum at c_1 and c_2 respectively,

$$c_1, c_2 \in (0, 1)$$

$$\text{Then, } f(c_1) = g(c_2)$$

$$\text{Let } h(x) = f(x) - g(x)$$

$$\text{Then, } h(c_1) = f(c_1) - g(c_1) > 0$$

$$\text{and } h(c_2) = f(c_2) - g(c_2) < 0$$

 $\therefore h(x) = 0$ has at least one root in (c_1, c_2)

$$C \in (c_1, c_2) \text{ i.e. for } f(c) = g(c)$$

which shows that (a) and (d) are correct.

$$23. (a,d) f(x) = \begin{cases} \frac{x}{|x|} g(x), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

$$= \begin{cases} -g(x), & x < 0 \\ 0, & x = 0 \\ g(x), & x > 0 \end{cases}$$

$$f'(x) = \begin{cases} -g'(x), & x < 0 \\ 0, & x = 0 \\ g'(x), & x > 0 \end{cases}$$

$$\therefore Lf'(0) = -g'(0) = 0$$

$$Rf'(0) = g'(0) = 0$$

 $\therefore f$ is differentiable at $x = 0$

$$h(x) = e^{|x|} = \begin{cases} e^{-x}, & x < 0 \\ e^x, & x \geq 0 \end{cases}$$

$$h'(x) = \begin{cases} -e^{-x}, & x < 0 \\ e^x, & x \geq 0 \end{cases} \Rightarrow Lh'(0) = -1, Rh'(0) = 1$$

 $\therefore h$ is not differentiable at $x = 0$

$$f \circ h(x) = f(h(x)) = g(e^{|x|}) \text{ as } e^{|x|} > 0$$

$$= \begin{cases} g(e^{-x}) & \text{if } x < 0 \\ g(1) & \text{if } x = 0 \\ g(e^x) & \text{if } x > 0 \end{cases}$$

$$f'[h(x)] = \begin{cases} -g'(e^{-x})e^{-x} & , x < 0 \\ 0 & , x = 0 \\ g'(e^x)e^x & , x > 0 \end{cases}$$

$$Lf'(h(0)) = -g'(1), Rf'(h(0)) = g'(1)$$

$$\therefore g'(1) \neq 0, \therefore Lf'(h(0)) \neq Rf'(h(0))$$

$$\therefore f \circ h \text{ is not differentiable at } x = 0.$$

$$hof(x) = \begin{cases} e^{|f(x)|} & , x \neq 0 \\ 1 & , x = 0 \end{cases}$$

$$Lh'(f(0)) = \lim_{k \rightarrow 0} \frac{h(f(0)) - h(f(0-k))}{k}$$

$$= \lim_{k \rightarrow 0} \frac{1 - e^{|g(-k)|}}{k} = \lim_{k \rightarrow 0} \frac{1 - e^{|g(-k)|}}{|g(-k)|} \times \frac{|g(-k)|}{k}$$

$$= 1 \times 0 = 0 \left(\because g'(0) = 0 \Rightarrow \lim_{k \rightarrow 0} \frac{g(-k)}{k} = \lim_{k \rightarrow 0} \frac{g(k)}{k} = 0 \right)$$

$$Rh'(f(0)) = \lim_{k \rightarrow 0} \frac{h(f(0+k)) - h(f(0))}{k}$$

$$= \lim_{k \rightarrow 0} \frac{e^{|g(k)|} - 1}{k} = \lim_{k \rightarrow 0} \frac{e^{|g(k)|} - 1}{|g(k)|} \times \frac{|g(k)|}{k} = 0$$

$$\therefore h \circ f \text{ is differentiable at } x = 0.$$

24. (a, b) $f(x) = a \cos^3(|x^3 - x|) + b|x| \sin(|x^3 + x|)$

(a) If $a = 0, b = 1$

$$\Rightarrow f(x) = |x| \sin |x^3 + x|$$

$$= x \sin(x^3 + x), x \in \mathbb{R}$$

$$\therefore f \text{ is differentiable every where.}$$

(b), (c) If $a = 1, b = 0 \Rightarrow f(x) = \cos^3(|x^3 - x|)$
 $= \cos^3(x^3 - x)$

$$\text{which is differentiable every where.}$$

(d) when $a = 1, b = 1, f(x) = \cos(x^3 - x) + x \sin(x^3 + x)$
 which is differentiable at $x = 1$

$$\therefore \text{Only a and b are the correct options.}$$

25. (b, c) $f(x) = [x^2 - 3]$ is discontinuous at all integral points in

$$\left[-\frac{1}{2}, 2\right]$$

$$\text{Which happens when } x = 1, \sqrt{2}, \sqrt{3}, 2$$

$$\therefore f \text{ is discontinuous exactly at four points in } \left[-\frac{1}{2}, 2\right]$$

$$\text{Also } g(x) = (|x| + |4x - 7|)f(x)$$

$$\text{Here } f \text{ is not differentiable at } x = 1, \sqrt{2}, \sqrt{3} \in \left[-\frac{1}{2}, 2\right]$$

$$\text{and } |x| + |4x - 7| \text{ is not differentiable at } 0 \text{ and } \frac{7}{4}$$

$$\text{But } f(x) = 0, \forall x \in [\sqrt{3}, 2]$$

$$\therefore g(x) \text{ becomes differentiable at } x = \frac{7}{4}$$

$$\text{Hence } g(x) \text{ is non-differentiable at four points i.e., } 0, 1, \sqrt{2}, \sqrt{3}$$

E. SUBJECTIVE PROBLEMS

1.

$$\begin{aligned} & \lim_{x \rightarrow a} \frac{\sqrt{a+2x} - \sqrt{3x}}{\sqrt{3a+x} - 2\sqrt{x}} \\ &= \lim_{x \rightarrow a} \frac{(\sqrt{a+2x} - \sqrt{3x})(\sqrt{a+2x} + \sqrt{3x})(\sqrt{3a+x} + 2\sqrt{x})}{(\sqrt{3a+x} - 2\sqrt{x})(\sqrt{3a+x} + 2\sqrt{x})(\sqrt{a+2x} + \sqrt{3x})} \\ &= \lim_{x \rightarrow a} \frac{(a+2x-3x)(\sqrt{3a+x} + 2\sqrt{x})}{(3a+x-4x)(\sqrt{a+2x} + \sqrt{3x})} \\ &= \lim_{x \rightarrow a} \frac{(a-x)(\sqrt{3a+x} + 2\sqrt{x})}{3(a-x)(\sqrt{a+2x} + \sqrt{3x})} \\ &= \lim_{x \rightarrow a} \frac{(\sqrt{3a+x} + 2\sqrt{x})}{3(\sqrt{a+2x} + \sqrt{3x})} = \frac{\sqrt{3a+a} + 2\sqrt{a}}{3(\sqrt{a+2a} + \sqrt{3a})} \\ &= \frac{4\sqrt{a}}{3 \times 2\sqrt{3a}} = \frac{2}{3\sqrt{3}} \end{aligned}$$

NOTE : The given limit is of the form $\frac{0}{0}$. Hence limit of the function can also be find out by using L' Hospital's Rule.

2.

$$f(x) = \int \frac{2 \sin x - \sin 2x}{x^3} dx, x \neq 0$$

$$\therefore f'(x) = \frac{2 \sin x - \sin 2x}{x^3}, x \neq 0$$

$$\begin{aligned} \therefore \lim_{x \rightarrow 0} f'(x) &= \lim_{x \rightarrow 0} \frac{2 \sin x - \sin 2x}{x^3} \\ &= \lim_{x \rightarrow 0} \frac{2 \sin x(1 - \cos x)(1 + \cos x)}{x^3(1 + \cos x)} \\ &= \lim_{x \rightarrow 0} 2 \cdot \frac{\sin^3 x}{x^3} \cdot \frac{1}{1 + \cos x} \\ &= 2 \times (1)^3 \times \frac{1}{2} = 1 \end{aligned}$$

3.

$$\begin{aligned} & \lim_{h \rightarrow 0} \frac{(a+h)^2 \sin(a+h) - a^2 \sin a}{h} \\ &= \lim_{h \rightarrow 0} \frac{a^2[\sin(a+h) - \sin a] + 2ah \sin(a+h) + h^2 \sin(a+h)}{h} \\ &= \lim_{h \rightarrow 0} \frac{a^2 \left[2 \cos \left(a + \frac{h}{2} \right) \sin \frac{h}{2} \right] + 2a \sin(a+h)}{2 \times \frac{h}{2}} + h \sin(a+h) \\ &= a^2 \cos a + 2a \sin a \end{aligned}$$

4.

$$\text{As } f(x) \text{ is continuous at } x = 0, \text{ we have}$$

$$\text{LHL} = \text{RHL} = f(0)$$

$$\Rightarrow \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} f(0+h) = f(0)$$

$$\Rightarrow f(0) + \lim_{h \rightarrow 0} f(-h) = f(0) + \lim_{h \rightarrow 0} f(h) = f(0)$$

$$[\text{Using the given property } f(x+y) = f(x) + f(y)]$$

$$\Rightarrow \lim_{h \rightarrow 0} f(-h) = \lim_{h \rightarrow 0} f(h) = 0 \quad \dots (1)$$

Limits, Continuity and Differentiability

Now let $x = a$ be any arbitrary point then at $x = a$,

$$\begin{aligned} \text{LHL} &= \lim_{h \rightarrow 0} f(a-h) = \lim_{h \rightarrow 0} [f(a) + f(-h)] \\ &= f(a) + \lim_{h \rightarrow 0} f(-h) = f(a) \quad [\text{Using, } f(x+y) = f(x) + f(y)] \\ &= f(a) + \lim_{h \rightarrow 0} f(-h) = f(a) \quad [\text{using eq}^n(1)] \end{aligned}$$

$$\text{Similarly, R.H.L.} = \lim_{h \rightarrow 0} f(a+h) = f(a)$$

Thus, we get

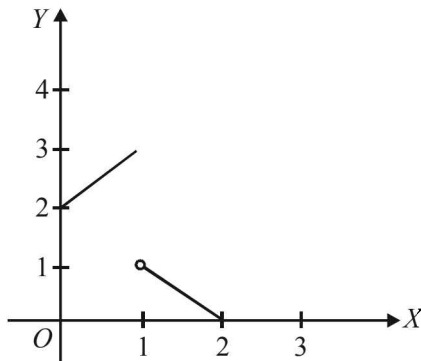
$$\lim_{h \rightarrow 0} f(a-h) = \lim_{h \rightarrow 0} f(a+h) = f(a)$$

$\Rightarrow f$ is continuous at $x = a$. But a is any arbitrary point

$\therefore f$ is continuous $\forall x \in R$.

$$\begin{aligned} 5. \quad \lim_{x \rightarrow 0} \frac{2^x - 1}{\sqrt{1+x} - 1} &= \lim_{x \rightarrow 0} \frac{2^x - 1}{\sqrt{1+x} - 1} \times \frac{\sqrt{1+x} + 1}{\sqrt{1+x} + 1} \\ &= \lim_{x \rightarrow 0} \frac{(2^x - 1)(\sqrt{1+x} + 1)}{1 + x - 1} \\ &= \lim_{x \rightarrow 0} \frac{2^x - 1}{x} \cdot \lim_{x \rightarrow 0} (\sqrt{1+x} + 1) \\ &= \ln 2 \cdot (1 + 1) = 2 \ln 2. \end{aligned}$$

6. Graph of $f(f(x))$ is



Clearly from graph $f(f(x))$ is discontinuous at $x = 1$ and 2.

$$7. \quad \text{We have } f(x) = \frac{x^2}{2}, 0 \leq x < 1 = 2x^2 - 3x + \frac{3}{2}, 1 \leq x \leq 2$$

Here $f(x)$ is continuous everywhere except possibly at $x = 1$

$$\Rightarrow \text{At } x = 1, Lf' = \frac{2}{2} \times 1 = 1; Rf' = 4 \times 1 - 3 = 1$$

$\Rightarrow f$ is differentiable and hence continuous at $x = 1$

$\therefore f(x)$ is continuous on $[0, 2]$

$$f'(x) = x, 0 \leq x < 1$$

$$= 4x - 3, 1 \leq x \leq 2$$

At $x = 1$,

$$\lim_{x \rightarrow 1^-} f'(x) = \lim_{h \rightarrow 0} f'(1-h) = \lim_{h \rightarrow 0} (1-h) = 1$$

$$\lim_{x \rightarrow 1^+} f'(x) = \lim_{h \rightarrow 0} f'(1+h) = \lim_{h \rightarrow 0} 4(1+h) - 3 = 1$$

$$f'(1) = 4 - 3 = 1$$

$\therefore f'$ is continuous at $x = 1$

$\therefore f'$ is continuous on $[0, 2]$

$$f''(x) = \begin{cases} 2, & 0 \leq x < 1 \\ 4, & 1 \leq x \leq 2 \end{cases}$$

Clearly $f''(x)$ is discontinuous at $x = 1$,

$\therefore f''(x)$ is discontinuous on $[0, 2]$.

$$\text{Given } f(x) = x^3 - x^2 + x + 1$$

$$\therefore f'(x) = 3x^2 - 2x + 1 = 3 \left(x^2 - \frac{2}{3}x + \frac{1}{3} \right)$$

$$= 3 \left[\left(x - \frac{1}{3} \right)^2 - \frac{1}{9} + \frac{1}{3} \right]$$

$$= 3 \left[\left(x - \frac{1}{3} \right)^2 + \frac{2}{9} \right] > 0 \forall x \in R.$$

Hence $f(x)$ is an increasing function of x for all real values of x .

Now $\max [f(t) : 0 \leq t \leq x]$ means the greatest value of

$f(t)$ in $0 \leq t \leq x$ which is obtained at $t = x$, since $f(t)$ is increasing for all t .

$$\therefore \max. [f(t) : 0 \leq t \leq x] = x^3 - x^2 + x + 1$$

Hence the function g is defined as follows :

$$g(x) = x^3 - x^2 + x + 1 \quad \text{when } 0 \leq x \leq 1$$

$$= 3 - x \quad \text{when } 1 < x \leq 2$$

Now it is sufficient to discuss the continuity and differentiability of $g(x)$ at $x = 1$. Since for all other values of x , $g(x)$ is clearly continuous and differentiable, being a polynomial function of x .

We have, $g(1) = 2$

$$g(1-0) = \lim_{h \rightarrow 0} [(1-h)^3 - (1-h)^2 + (1-h) + 1] = 2$$

$$g(1+0) = \lim_{h \rightarrow 0} [3 - (1+h)] = 2$$

Hence $g(x)$ is continuous at $x = 1$

Now,

$$Lg'(1) = \lim_{h \rightarrow 0} \frac{[(1-h)^3 - (1-h)^2 + (1-h) + 1] - 2}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{1 - 3h + 3h^2 - h^3 - 1 + 2h - h^2 + 1 - h + 1 - 2}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{-2h + 2h^2 - h^3}{-h} = \lim_{h \rightarrow 0} [2 - 2h + h^2] = 2$$

$$Rg'(1) = \lim_{h \rightarrow 0} \frac{[3 - (1+h) - 2]}{h} = \lim_{h \rightarrow 0} \frac{-h}{h} = -1$$

Since $Lg'(1) \neq Rg'(1)$, the function $g(x)$ is not differentiable at $x = 1$

Hence $g(x)$ is continuous on $(0, 2)$. It is also differentiable on $(0, 2)$ except at $x = 1$.

We have $f(x) = -1, -2 \leq x \leq 0$

$$= x - 1, 0 < x \leq 2$$

$$\text{and } g(x) = f(|x| + |f(x)|)$$

Hence $g(x)$ involves $|x|$ and $|x-1|$ or $|-1| = 1$

Therefore we should divide the given interval $(-2, 2)$ into the following intervals.

8.

9.

I_1	I_2	I_3
$[-2, 2] = [-2, 0)$	$[0, 1)$	$[1, 2]$
$x = -ve$	$+ve$	$+ve$
$ x = -x$	x	x
$f(x) = -1$	$x - 1$	$x - 1$
$f(x) = -1$	$= x - 1$	$= x - 1$
$ f(x) = -1 $	$ x - 1 $	$ x - 1 $
$= 1$	$= -(x - 1)$	$= x - 1$

$\therefore$ Using above we get

$$\begin{aligned}
 g(x) &= f(|x|) + |f(x)| \\
 &= -1 + 1 = 0 \text{ in } I_1 \\
 &= x - 1 - (x - 1) = 0 \text{ in } I_2 \\
 &= x - 1 + x - 1 = 2(x - 1) \text{ in } I_3
 \end{aligned}$$

Hence $g(x)$ is defined as follows :

$$g(x) = \begin{cases} 0, & -2 \leq x < 1 \\ 2(x - 1), & 1 \leq x \leq 2 \end{cases}$$

$$Lg'(1) = 0; Rg'(1) = 2 \text{ (not equal)}$$

Hence $g(x)$ is not differentiable at $x = 1$.

10. Let $h(x) = f(x) + g(x)$ be continuous.

$$\text{Then, } g(x) = h(x) - f(x)$$

Now, $h(x)$ and $f(x)$ both are continuous functions.

$\therefore h(x) - f(x)$ must also be continuous. But it is a contradiction as given that $g(x)$ is discontinuous. Therefore our assumption that $f(x) + g(x)$ a continuous function is wrong and hence $f(x) + g(x)$ is discontinuous.

11. Given that $f(x)$ is a function satisfying

$$f(-x) = f(x), \forall x \in R \quad \dots(1)$$

Also $f'(0)$ exists

$$\Rightarrow f'(0) = Rf'(0) = Lf'(0)$$

$$\text{Now, } Rf'(0) = f'(0)$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = f'(0)$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = f'(0)$$

$$\text{Again } Lf'(0) = f'(0) \quad \dots(2)$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} = f'(0)$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{f(-h) - f(0)}{-h} = f'(0)$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = f'(0) \quad \dots(3)$$

[Using eq. (1)]

From equations (2) and (3), we get

$$\Rightarrow f'(0) = 0$$

12. Given that,

$$f(x) = \begin{cases} x + a\sqrt{2} \sin x & , 0 \leq x < \frac{\pi}{4} \\ 2x \cot x + b & , \frac{\pi}{4} \leq x \leq \frac{\pi}{2} \\ a \cos 2x - b \sin x & , \frac{\pi}{2} < x \leq \pi \end{cases}$$

is continuous for $0 \leq x \leq \pi$.

$$\therefore f(x) \text{ must be continuous at } x = \frac{\pi}{4} \text{ and } x = \frac{\pi}{2}$$

$$\lim_{x \rightarrow \left(\frac{\pi}{4}\right)^-} f(x) = f\left(\frac{\pi}{4}\right)$$

$$\Rightarrow \lim_{h \rightarrow 0} f\left(\frac{\pi}{4} - h\right) = \frac{2\pi}{4} \cot \frac{\pi}{4} + b$$

$$\Rightarrow \lim_{h \rightarrow 0} \left(\frac{\pi}{4} - h\right) + a\sqrt{2} \sin\left(\frac{\pi}{4} - h\right) = \frac{\pi}{2} + b$$

$$\Rightarrow \frac{\pi}{4} + a = \frac{\pi}{2} + b$$

$$\Rightarrow a - b = \frac{\pi}{4} \quad \dots(1)$$

$$\text{Also, } \lim_{x \rightarrow \left(\frac{\pi}{2}\right)^+} f(x) = f\left(\frac{\pi}{2}\right)$$

$$\Rightarrow \lim_{h \rightarrow 0} f\left(\frac{\pi}{2} + h\right) = 2 \cdot \frac{\pi}{2} \cot \frac{\pi}{2} + b$$

$$\Rightarrow \lim_{h \rightarrow 0} a \cos 2\left(\frac{\pi}{2} + h\right) - b \sin\left(\frac{\pi}{2} + h\right) = b$$

$$\Rightarrow a \cos \pi - b \sin \frac{\pi}{2} = b \Rightarrow -a - b = b$$

$$\Rightarrow a + 2b = 0 \quad \dots(2)$$

Solving (1) and (2), we get $a = \frac{\pi}{6}$ and $b = \frac{-\pi}{12}$.

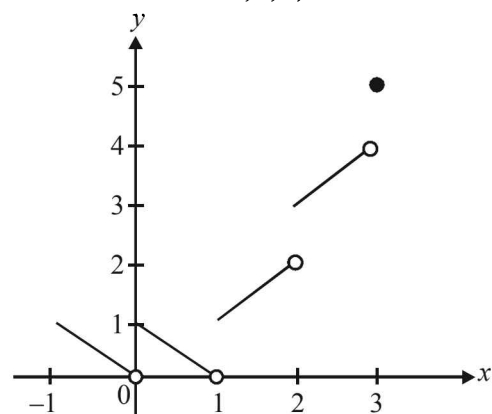
We have, $[x] + |1 - x|$, $-1 \leq x \leq 3$

NOTE THIS STEP

$$\text{or } y = \begin{cases} -1 + 1 - x & , -1 \leq x < 0 \\ 0 + 1 - x & , 0 \leq x < 1 \\ 1 - 1 + x & , 1 \leq x < 2 \\ 2 - 1 + x & , 2 \leq x < 3 \\ 3 - 1 + x & , x = 3 \end{cases}$$

$$\text{or } y = \begin{cases} -x & , -1 \leq x < 0 \\ 1 - x & , 0 \leq x < 1 \\ x & , 1 \leq x < 2 \\ 1 + x & , 2 \leq x < 3 \\ 2 + x & , x = 3 \end{cases}$$

From graph we can say that given functions is not differentiable at $x = 0, 1, 2, 3$.



14. We are given that,

$$f(x) = \begin{cases} \frac{1 - \cos 4x}{x^2}, & x < 0 \\ a, & x = 0 \\ \frac{\sqrt{x}}{\sqrt{16 + \sqrt{x}} - 4}, & x > 0 \end{cases}$$

Here L.H.L at $(x=0)$

$$= \lim_{h \rightarrow 0} \frac{1 - \cos 4(0-h)}{(0-h)^2} = \lim_{h \rightarrow 0} \frac{1 - \cos 4h}{h^2}$$

$$= \lim_{h \rightarrow 0} \frac{2 \sin^2 2h}{4h^2} \cdot 4 = 8$$

R.H.L at $(x=0)$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{0+h}}{\sqrt{16+\sqrt{0+h}}-4} = \lim_{h \rightarrow 0} \frac{\sqrt{h}(\sqrt{16+h}+4)}{16+\sqrt{h}-16}$$

$$= \lim_{h \rightarrow 0} \sqrt{16+\sqrt{h}}+4 = \sqrt{16}+4 = 8$$

For continuity of function $f(x)$, we must have

$$\text{L.H.L.} = \text{R.H.L.} = f(0)$$

$$\Rightarrow f(0) = 8 \Rightarrow a = 8$$

15. We are given

$$f(x+y) = f(x)f(y), \forall x, y \in R$$

$$f(x) \neq 0, \text{ for any } x$$

$$f \text{ is differentiable at } x=0, f'(0) = 2$$

To prove that $f'(x) = 2f(x)$, $\forall x \in R$ and to find $f(x)$.

We have for $x=y=0$

$$f(0+0) = f(0)f(0)$$

$$\Rightarrow f(0) = [f(0)]^2 \Rightarrow f(0) = 1$$

Again $f'(0) \neq 2$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = 2 \Rightarrow \lim_{h \rightarrow 0} \frac{f(0)f(h) - f(0)}{h} = 2$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{f(0)[f(h)-1]}{h} = 2$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{f(h)-1}{h} = 2 \quad \dots(1) \quad [\text{Using } f(0)=1]$$

$$\text{Now, } f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(x)f(h) - f(x)}{h} = \lim_{h \rightarrow 0} f(x) \left(\frac{f(h)-1}{h} \right)$$

$$= f(x) \lim_{h \rightarrow 0} \left[\frac{f(h)-1}{h} \right]$$

$$= f(x) \cdot 2 \quad [\text{Using eq. (1)}]$$

$$= 2f(x)$$

$$\text{Also, } \frac{f'(x)}{f(x)} = 2$$

Integrating on both sides with respect to x , we get

$$\log |f(x)| = 2x + C$$

$$\text{At } x=0, \log f(0) = C \Rightarrow C = \log 1 = 0$$

$$\therefore \log |f(x)| = 2x \Rightarrow f(x) = e^{2x}$$

16.

$$\lim_{x \rightarrow 0} \left\{ \tan\left(\frac{\pi}{4}\right) + x \right\}^{\frac{1}{x}} = e^{\lim_{x \rightarrow 0} \log \left\{ \tan\left(\frac{\pi}{4}\right) + x \right\}^{\frac{1}{x}}}$$

$$[\text{Using } \lim_{x \rightarrow a} f(x) = e^{\lim_{x \rightarrow 0} \log f(x)}]$$

$$= e^{\lim_{x \rightarrow 0} \frac{\log \tan\left(\frac{\pi}{4} + x\right)}{x}} \quad \left[\frac{0}{0} \text{ form} \right]$$

$$= e^{\lim_{x \rightarrow 0} \left[\frac{\sec^2\left(\frac{\pi}{4} + x\right)}{\tan\left(\frac{\pi}{4}\right) + x} \right]}$$

[Using L' Hospital's rule]

$$= e^{\frac{2}{1}} = e^2$$

17.

$$\text{Given that, } f(x) = \begin{cases} (1 + |\sin x|)^{\frac{a}{|\sin x|}} & -\frac{\pi}{6} < x < 0 \\ b & x = 0 \\ e^{\frac{\tan 2x}{\tan 3x}} & 0 < x < \frac{\pi}{6} \end{cases}$$

is continuous at $x=0$

$$\therefore \lim_{h \rightarrow 0} f(0-h) = f(0) = \lim_{h \rightarrow 0} f(0+h)$$

We have,

$$\lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} [1 + |\sin(-h)|]^{\frac{a}{|\sin(-h)|}}$$

$$= \lim_{h \rightarrow 0} [1 + \sin h]^{\frac{a}{\sin h}}$$

$$\lim_{h \rightarrow 0} \frac{a}{\sin h} \log(1 + \sin h) = e^a$$

$$\text{and } f(0) = b$$

$$\therefore e^a = b \quad \dots(1)$$

$$\text{Also } \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} e^{\frac{\tan 2h}{\tan 3h}}$$

$$= e^{\lim_{h \rightarrow 0} \frac{\tan 2h}{2h} \times \frac{3h}{\tan 3h} \times \frac{2}{3}} = e^{\frac{2}{3}}$$

$$\therefore e^{\frac{2}{3}} = b \quad \dots(2)$$

From (1) and (2)

$$e^a = b = e^{\frac{2}{3}} \Rightarrow a = \frac{2}{3} \text{ and } b = e^{\frac{2}{3}}$$

18.

$$f\left(\frac{x+y}{2}\right) = \frac{f(x) + f(y)}{2} \quad \dots(1)$$

Putting $y=0$ and $f(0)=1$ in (1), we get

$$f\left(\frac{x}{2}\right) = \frac{1}{2}[f(x) + 1]$$

$$\therefore f(x) = 2f\left(\frac{x}{2}\right) - 1 \quad \dots(2)$$

$$\text{Now, } f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\begin{aligned}
 &= \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{f(2x) + f(2h)}{2} - f(x) \right], \text{ by (1)} \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{(2f(x) - 1) + (2f(h) - 1)}{2} - f(x) \right], \\
 &\quad \text{by (2)} \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} [f(h) - 1] \\
 &= \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = f'(0) = -1
 \end{aligned}$$

Hence $f'(x) = -1$, integrating, we get

$f(x) = -x + c$. Putting $x = 0$, we get

$f(0) = c = 1$ by (1) $\therefore f(x) = 1 - x$

$f(2) = 1 - 2 = -1$

19. By the given definition it is clear that the function f is continuous and differentiable at all points except possible at $x = 1$ and $x = 2$.

Continuity at $x = 1$

$$\text{L.H.L.} = \lim_{h \rightarrow 0} [1 - (1 - h)] = \lim_{h \rightarrow 0} h = 0$$

$$\text{R.H.L.} = \lim_{h \rightarrow 0} [1 - (1 + h)][2 - (1 + h)]$$

$$= \lim_{h \rightarrow 0} \{-h(1 - h)\} = 0$$

Also, $f(1) = 0$

$\therefore \text{L.H.L.} = \text{R.H.L.} = f(1) = 0$

Therefore, f is continuous at $x = 1$

Now, differentiability at $x = 1$

$$\begin{aligned}
 Lf'(1) &= \lim_{h \rightarrow 0} \frac{f(1 - h) - f(1)}{-h}, h > 0 \\
 &= \lim_{h \rightarrow 0} \frac{(1 - (1 - h)) - 0}{-h} = \lim_{h \rightarrow 0} \left(\frac{h}{-h} \right) = -1
 \end{aligned}$$

$$\begin{aligned}
 \text{and } Rf'(1) &= \lim_{h \rightarrow 0} \frac{f((1 + h) - f(1))}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\{1 - (1 + h)\} \{2 - (1 + h)\} - 0}{h} \\
 &= \lim_{h \rightarrow 0} \frac{-h(1 - h)}{h} = \lim_{h \rightarrow 0} (h - 1) = -1
 \end{aligned}$$

Since $Lf'(1) = Rf'(1)$

Hence, f is differentiable at $x = 1$

Continuous at $x = 2$

$$\text{L.H.L.} = \lim_{h \rightarrow 0} [1 - (2 - h)][2 - (2 - h)]$$

$$\lim_{h \rightarrow 0} \{(-1 + h)\} \{h\} = 0$$

$$\text{and R.H.L.} = \lim_{h \rightarrow 0} [3 - (2 + h)] = \lim_{h \rightarrow 0} (1 - h) = 1$$

Since $\text{L.H.L.} \neq \text{R.H.L.}$, therefore f is not continuous at $x = 2$. As such f cannot be differentiable at $x = 2$. Hence f is continuous and differentiable at all points except at $x = 2$.

20. Given that, $F(x) = \int_0^x f(t) dt$

NOTE THIS STEP

$$\therefore F'(x) = f(x) \cdot 1 - f(0) \cdot 0$$

[Using Leibnitz theorem]

$$\Rightarrow F'(x) = f(x) \dots (1), \forall x \geq 0$$

$$\text{Also } F(0) = \int_0^0 f(t) dt = 0$$

But given that $f(x) \leq cF(x), \forall x \geq 0$

$\therefore$ We get $f(0) \leq cF(0) = 0$

$$\therefore f(0) \leq 0 \dots (2)$$

But ATQ $f(x)$ is non-negative continuous function on $[0, \infty)$

$$\therefore f(x) \geq 0$$

$$\therefore f(0) \geq 0 \dots (3)$$

$\therefore$ From (2) and (3) $f(0) = 0$

Again $f(x) \leq cF(x) \forall x \geq 0$, we get

$$f(x) - cF(x) \leq 0$$

$$\Rightarrow F'(x) - cF(x) \leq 0, \forall x \geq 0 \text{ [Using equation (1)]}$$

$$e^{cx} F'(x) - ce^{-cx} F(x) \leq 0$$

[Multiplying both sides by e^{-cx} (I.F.) and keeping in mind that $e^{-cx} > 0, \forall x$]

$$\Rightarrow \frac{d}{dx} [e^{-cx} F(x)] \leq 0$$

$\Rightarrow g(x) = e^{-cx} F(x)$ is a decreasing function on $[0, \infty)$.

That is $g(x) \leq g(0)$ for all $x \geq 0$

But $g(0) = F(0) = 0$

$$\therefore g(x) \leq 0, \forall x \geq 0$$

$$\Rightarrow e^{-cx} F(x) \leq 0, \forall x \geq 0$$

$$\Rightarrow F(x) \leq 0, \forall x \geq 0$$

$$\therefore f(x) \leq cF(x) \leq 0, \forall x \geq 0$$

[$\therefore c > 0$ and using $f(x) \leq cF(x)$]

$$\Rightarrow f(x) \leq 0, \forall x \geq 0$$

But given $f(x) \geq 0$

$$\Rightarrow f(x) = 0, \forall x \geq 0.$$

21. (I) g is continuous at α and

$$f(x) - f(\alpha) = g(x)(x - \alpha), \forall x \in R$$

$\Rightarrow$ Since g is continuous at $x = \alpha$

$$\text{and } g(x) = \frac{f(x) - f(\alpha)}{x - \alpha}$$

We should have, $\lim_{x \rightarrow \alpha} g(x) = g(\alpha)$

$$\Rightarrow \lim_{x \rightarrow \alpha} \frac{f(x) - f(\alpha)}{x - \alpha} = g(\alpha) \Rightarrow f'(\alpha) = g(\alpha)$$

$\Rightarrow f'(\alpha)$ exists and is equal to $g(\alpha)$.

- (II) $\therefore f(x)$ is differentiable at $x = \alpha$

$$\therefore \lim_{x \rightarrow \alpha} \frac{f(x) - f(\alpha)}{x - \alpha} = f'(\alpha)$$

exists and is finite.

Limits, Continuity and Differentiability

Let us define,

$$g(x) = \begin{cases} \frac{f(x) - f(\alpha)}{x - \alpha}, & x \neq \alpha \\ f'(\alpha), & x = \alpha \end{cases}$$

Then, $f(x) - f(\alpha) = (x - \alpha)g(x)$, $\forall x \neq \alpha$.

Now for continuity of $g(x)$ at $x = \alpha$

$$\lim_{x \rightarrow \alpha} g(x) = \lim_{x \rightarrow \alpha} \frac{f(x) - f(\alpha)}{x - \alpha} = f'(\alpha) = g(\alpha)$$

$\therefore g$ is continuous at $x = \alpha$.

22.

Given that

$$f(x) = \begin{cases} x+a, & \text{if } x < 0 \\ |x-1|, & \text{if } x \geq 0 \end{cases} = \begin{cases} x+a, & \text{if } x < 0 \\ 1-x, & \text{if } 0 \leq x < 1 \\ x-1, & \text{if } x \geq 1 \end{cases}$$

$$\text{and } g(x) = \begin{cases} (x+1), & \text{if } x < 0 \\ (x-1)^2 + b, & \text{if } x \geq 0 \end{cases}$$

where $a, b \geq 0$

Then $(g \circ f)(x) = g[f(x)]$

NOTE THIS STEP

$$= \begin{cases} f(x)+1, & \text{if } f(x) < 0 \\ [f(x)-1]^2 + b, & \text{if } f(x) \geq 0 \end{cases}$$

(Using definition of $g(x)$)

Now, $f(x) < 0$ when $x+a < 0$ i.e. $x < -a$

$f(x) = 0$ when $x = -a$ or $x = 1$

$f(x) > 0$ when $-a < x < 1$ or $x > 1$

$$g(f(x)) = \begin{cases} f(x)+1, & \text{if } x < -a \\ [f(x)-1]^2 + b, & \text{if } x = -a \text{ or } x = 1 \\ [f(x)-1]^2 + b, & \text{if } -a < x < 1 \\ [f(x)-1]^2 + b, & \text{if } 0 \leq x < 1 \\ [f(x)-1]^2 + b, & \text{if } x > 1 \end{cases}$$

[Keeping in mind that $x=0$ and 1 are also the breaking points because of definition of $f(x)$]

$$\therefore g[f(x)] = \begin{cases} x+a+1, & \text{if } x < -a \\ (x+a-1)^2 + b, & \text{if } -a \leq x < 0 \\ (1+x-1)^2 + b, & \text{if } 0 \leq x \leq 1 \\ (x-1-1)^2 + b, & \text{if } x > 1 \end{cases}$$

Substituting the value of $f(x)$ under different conditions).

$$\therefore g[f(x)] = \begin{cases} x+a+1, & \text{if } x < -a \\ (x+a-1)^2 + b, & \text{if } -a \leq x < 0 = F(x) \text{ (say)} \\ x^2 + b, & \text{if } 0 \leq x \leq 1 \\ (x-2)^2 + b, & \text{if } x > 1 \end{cases}$$

Now given that $g \circ f(x) \equiv F(x)$ is continuous for all real numbers, therefore it will be continuous at $-a$

$\Rightarrow$ L.H.S = R.H.L = $f(-a)$

$$\lim_{h \rightarrow 0} F(-a-h) = \lim_{h \rightarrow 0} F(-a+h) = f(-a)$$

$$\text{Now, } \lim_{h \rightarrow 0} F(-a-h) = \lim_{h \rightarrow 0} (-a-h+a+1) = 1$$

$$\lim_{h \rightarrow 0} F(-a+h) = \lim_{h \rightarrow 0} (-a+h+a-1)^2 + b = 1+b$$

$$F(-a) = 1+b$$

Thus we should have $1 = 1+b \Rightarrow b = 0$.

Again for continuity at $x = 0$

$$\text{L.H.L} = f(0)$$

$$\Rightarrow \lim_{h \rightarrow 0} f(0-h) = f(0)$$

$$\Rightarrow \lim_{h \rightarrow 0} f(-h+a-1)^2 + b = b \Rightarrow (a-1)^2 = 0 \Rightarrow a = 1$$

For $a = 1$ and $b = 0$, $g \circ f$ becomes

$$g \circ f(x) = \begin{cases} x+2, & x < -1 \\ x^2, & -1 \leq x \leq 1 \\ (x-2)^2, & x > 1 \end{cases}$$

Now to check differentiability of $g \circ f(x)$ at $x = 0$

We see, $g \circ f(x) = x^2 = F(x)$

$\Rightarrow F'(x) = 2x$ which exists clearly at $x = 0$.

$g \circ f$ is differentiable at $x = 0$

Given that $f: [-2a, 2a] \rightarrow \mathbb{R}$

f is an odd function.

Lf' at $x = a$ is 0.

$$\Rightarrow \lim_{h \rightarrow 0} \frac{f(a-h) - f(a)}{-h} = 0$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{f(a-h) - f(a)}{h} = 0 \quad \dots(1)$$

To find Lf' at $x = -a$ which is given by

$$\lim_{h \rightarrow 0} \frac{f(-a-h) - f(-a)}{-h} = \lim_{h \rightarrow 0} \frac{-f(a+h) + f(a)}{-h} \quad [\because f(-x) = -f(x)]$$

$$= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Again for $x \in [a, 2a]$

$$f(x) = f(2a-x)$$

$$\therefore f(a+h) = f(2a-a-h) = f(a-h)$$

Substituting this values in last expression we get

$$Lf'(-a) = \lim_{h \rightarrow 0} \frac{f(a-h) - f(a)}{h} = 0 \quad [\text{Using equation (1)}]$$

Hence $Lf'(-a) = 0$

To find,

$$\lim_{n \rightarrow \infty} \left[(n+1) \frac{2}{\pi} \cos^{-1} \left(\frac{1}{n} \right) - n \right]$$

$$= \lim_{n \rightarrow \infty} n \left[\left(1 + \frac{1}{n} \right) \frac{2}{\pi} \cos^{-1} \left(\frac{1}{n} \right) - 1 \right] = \lim_{n \rightarrow \infty} n f \left(\frac{1}{n} \right)$$

where $f(x) = \left[(1+x) \frac{2}{\pi} \cos^{-1} x - 1 \right]$ such that

$$f(0) = \left[(1+0) \frac{2}{\pi} \cos^{-1} 0 - 1 \right] = \frac{2}{\pi} \cdot \frac{\pi}{2} - 1 = 0$$

$\therefore$ Using given relation as $\lim_{n \rightarrow \infty} n f\left(\frac{1}{n}\right) = f'(0)$

then given limit becomes

$$= f'(0) = \frac{d}{dx} \left[(1+x) \frac{2}{\pi} \cos^{-1} x - 1 \right] \Big|_{x=0}$$

$$= \frac{2}{\pi} \left[\cos^{-1} x - \frac{1-x}{\sqrt{1-x^2}} \right] \Big|_{x=0}$$

$$= \frac{2}{\pi} \left[\frac{\pi}{2} - 1 \right] = 1 - \frac{2}{\pi} = \frac{\pi-2}{\pi}$$

25.

Given that, $f(x)$ is differentiable at $x=0$.

Hence, $f(x)$ will also be continuous at $x=0$

$$\Rightarrow \lim_{h \rightarrow 0} f(0+h) = f(0) \Rightarrow \lim_{h \rightarrow 0} \frac{e^{\frac{ah}{2}} - 1}{h} = \frac{1}{2}$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{e^{\frac{ah}{2}} - 1}{\frac{ah}{2}} \times \frac{a}{2} = \frac{1}{2} \Rightarrow a = 1$$

Also differentiability of $f(x)$ at $x=0$, gives

$$Lf'(0) = Rf'(0)$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{b \sin^{-1} \left(\frac{c-h}{2} \right) - \frac{1}{2}}{-h} = \lim_{h \rightarrow 0} \frac{e^{\frac{ah}{2}} - 1 - \frac{1}{2}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2e^{\frac{ah}{2}} - 2 - h}{2h^2} \quad \left[\text{form } \frac{0}{0} \right]$$

$$\lim_{h \rightarrow 0} \frac{\sqrt{1 - \left(\frac{c-h}{2} \right)^2} \cdot \left(-\frac{1}{2} \right)}{-1} \quad [\text{Using L Hospital's rule}]$$

$$= \lim_{h \rightarrow 0} \frac{2e^{\frac{ah}{2}} \cdot \frac{a}{2} - 1}{4h} = \lim_{h \rightarrow 0} \frac{e^{\frac{h}{2}} - 1}{8 \left(\frac{h}{2} \right)} \quad [\text{Putting } a = 1]$$

$$\Rightarrow \frac{b}{\sqrt{1 - \frac{c^2}{4}}} = \frac{1}{8} \Rightarrow 4b = \sqrt{1 - \frac{c^2}{4}} \Rightarrow 16b^2 = \frac{4 - c^2}{4}$$

$$\Rightarrow 64b^2 = 4 - c^2 \quad \text{Hence proved.}$$

26.

Given that,

$$f(x-y) = f(x) \cdot g(y) - f(y) \cdot g(x) \quad \dots(i)$$

$$g(x-y) = g(x) \cdot g(y) + f(x)f(y) \quad \dots(ii)$$

In eqn. (i), putting $x=y$, we get

$$f(0) = f(x)g(x) - f(x)g(x) \Rightarrow f(0) = 0$$

Putting $y=0$, in eqn. (i), we get

$$f(x) = f(x)g(0) - f(0)g(x)$$

$$\Rightarrow f(x) = f(x)g(0) \quad [\text{using } f(0)=0]$$

$$\Rightarrow g(0) = 1$$

Putting $x=y$ in eqn. (ii), we get

$$g(0) = g(x)g(x) + f(x)f(x)$$

$$\Rightarrow 1 = [g(x)]^2 + [f(x)]^2 \quad [\text{using } g(0)=1]$$

$$\Rightarrow [g(x)]^2 = 1 - [f(x)]^2 \quad \dots(iii)$$

Clearly $g(x)$ will be differentiable only if $f(x)$ is differentiable.

$\therefore$ First we will check the differentiability of $f(x)$

Given that $Rf'(0)$ exists

$$\text{i.e., } \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} \text{ exists}$$

$$\text{i.e., } \lim_{h \rightarrow 0} \frac{f(0)g(-h) - f(-h)g(0)}{h} \text{ exists}$$

$$\text{i.e., } \lim_{h \rightarrow 0} \frac{-f(-h)}{h} \text{ exists (using } f(0)=0 \text{ and } g(0)=1)$$

Which can be written as,

$$\lim_{h \rightarrow 0} \frac{f(0) - f(-h)}{-h} = Lf'(0)$$

$$\Rightarrow Lf'(0) = Rf'(0)$$

$\therefore f$ is differentiable, at $x=0$

Differentiating equation (iii), we get

$$2g(x) \cdot g'(x) = -2f(x) \cdot f'(x)$$

For $x=0$

$$\Rightarrow g(0) \cdot g'(0) = -f(0)f'(0)$$

$$\Rightarrow g'(0) = 0 \quad [\text{Using } f(0)=0 \text{ and } g(0)=1]$$

F. Match the Following

1. (A) $\sin(\pi[x]) = 0, \forall x \in \mathbb{R}$

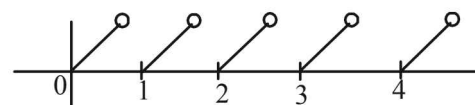
$\therefore$ Differentiable everywhere.

$\therefore$ (A) $\rightarrow$ (p)

- (B) $\sin(\pi(x - [x])) = f(x)$

We know that

$$x - [x] = \begin{cases} = x, & \text{if } 0 \leq x < 1 \\ = x - 1, & \text{if } 1 \leq x < 2 \\ = x - 2, & \text{if } 2 \leq x < 3 \end{cases}$$

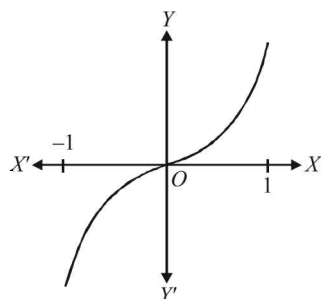


Its graph is, as shown in figure which is discontinuous at $\forall x \in \mathbb{Z}$. Clearly $x - [x]$ and hence $\sin(\pi(x - [x]))$ is not differentiable $\forall x \in \mathbb{Z}$

(B) $\rightarrow$ r

2. (A) $y = x|x| = \begin{cases} -x^2 & \text{if } x < 0 \\ x^2 & \text{if } x \geq 0 \end{cases}$

Graph is as follows :

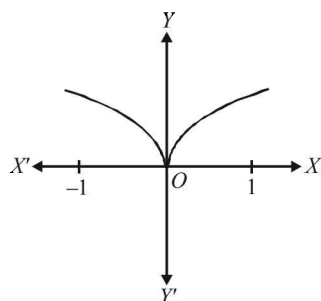


From graph $y = x|x|$ is continuous in $(-1, 1)$ (p)
 differentiable in $(-1, 1)$ (q)
 Strictly increasing in $(-1, 1)$. (r)

(B) $y = \sqrt{|x|} = \begin{cases} \sqrt{-x} & \text{if } x < 0 \\ \sqrt{x} & \text{if } x \geq 0 \end{cases}$
 $\Rightarrow y^2 = -x, x < 0$
 {where y can take only +ve values}

and $y^2 = x, x \geq 0$

$\therefore$ Graph is as follows :

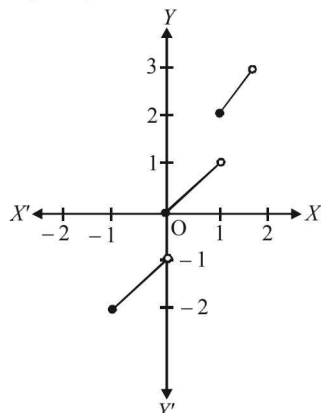


From graph $y = \sqrt{|x|}$ is continuous in $(-1, 1)$ (p)
 not differentiable at $x = 0$ (s)

(C) **NOTE THIS STEP**

$$y = x + [x] = \begin{cases} - & - & - \\ x-1, & -1 \leq x < 0 \\ x, & 0 \leq x < 1 \\ x+1, & 1 \leq x < 2 \\ - & - & - \end{cases}$$

$\therefore$ Graph of $y = x + [x]$ is as follows :

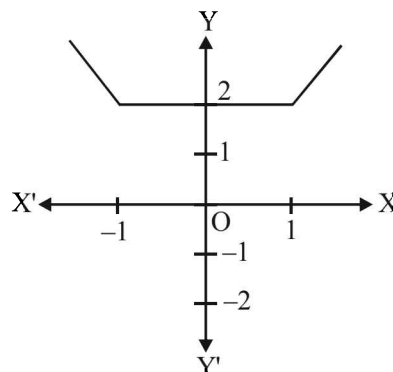


From graph, $y = x + [x]$ is neither continuous, nor differentiable at $x = 0$ and hence in $(-1, 1)$. (s)

Also it is strictly increasing in $(-1, 1)$ (r)

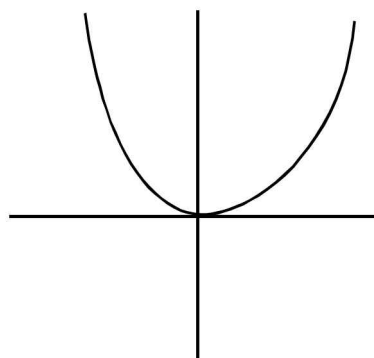
(D) $y = |x-1| + |x+1| = \begin{cases} -2x, & x < -1 \\ 2, & -1 \leq x < 1 \\ 2x, & x \geq 1 \end{cases}$

Graph of function is as follows :



From graph, $y = f(x)$ is continuous (p) and differentiable (q) in $(-1, 1)$ but not strictly increasing in $(-1, 1)$.

3. (d) $P(1): f_4(x) = \begin{cases} x^2, & x < 0 \\ e^{2x} - 1, & x \geq 0 \end{cases}$

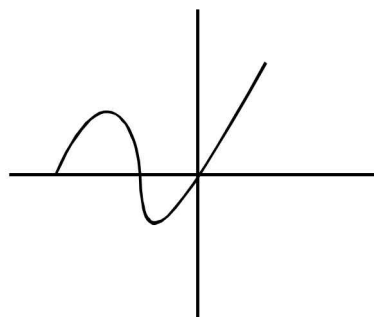


Range of $f_4 = [0, \infty)$

$\therefore f_4$ is onto

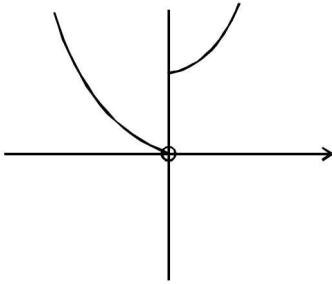
From graph f_4 is not one one.

$Q(3): f_3(x) = \begin{cases} \sin x, & x < 0 \\ x, & x \geq 0 \end{cases}$



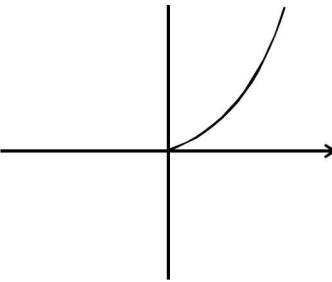
From graph f is differentiable but not one one.

$R(2): f_2 \circ f_1(x) = \begin{cases} x^2, & x < 0 \\ e^{2x}, & x \geq 0 \end{cases}$



From graph f_2 of f_1 is neither continuous nor one one.

$$S(4): f_2(x) = x^2, x \in [0, \infty)$$



It is continuous and one one.

I. Integer Value Correct Type

1. (6)

$$6 \int_1^x f(t) dt = 3xf(x) - x^3$$

Differentiating, we get $6f(x) = 3f(x) + 3xf'(x) - 3x^2$

$$\Rightarrow f'(x) - \frac{1}{x}f(x) = x$$

$$\text{I.F.} = \frac{1}{x}$$

$$\therefore \text{Solution is } f(x) \cdot \frac{1}{x} = \int 1 \cdot dx = x + c$$

$$\therefore f(x) = x^2 + cx$$

$$\text{But } f(1) = 2 \Rightarrow c = 1 \therefore f(x) = x^2 + x$$

$$\text{Hence } f(2) = 4 + 2 = 6$$

Note : Putting $x = 1$ in given integral equation, we get

$$f(1) = \frac{1}{3} \text{ while given } f(1) = 2.$$

$\therefore$ Data given in the question is inconsistent.

$$2. (2) \lim_{x \rightarrow 1} \left\{ \frac{-ax + \sin(x-1) + a}{x + \sin(x-1) - 1} \right\}^{\frac{1-x}{1-\sqrt{x}}} = \frac{1}{4}$$

$$\Rightarrow \lim_{x \rightarrow 1} \left\{ \frac{a(1-x) + \sin(x-1)}{(x-1) + \sin(x-1)} \right\}^{1+\sqrt{x}}$$

$$\Rightarrow \lim_{x \rightarrow 1} \left\{ \frac{-a + \frac{\sin(x-1)}{x-1}}{1 + \frac{\sin(x-1)}{x-1}} \right\}^{1+\sqrt{x}} \Rightarrow \left(\frac{-a+1}{2} \right)^2 = \frac{1}{4}$$

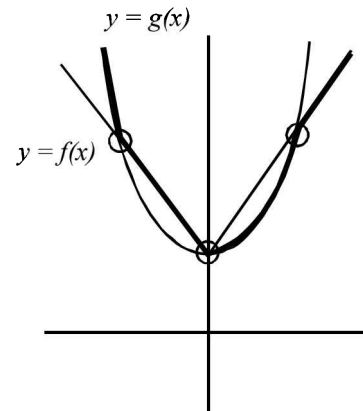
$$\Rightarrow a = 0 \text{ or } 2$$

$\therefore$ Largest value of a is 2.

3. (3)

$$f(x) = |x| + 1 = \begin{cases} x+1, & x \geq 0 \\ -x+1, & x < 0 \end{cases}$$

$$g(x) = x^2 + 1$$



From graph there are 3 points at which $h(x)$ is not differentiable.

$$4. (2) \lim_{\alpha \rightarrow 0} \frac{e^{\cos \alpha^n} - e}{\alpha^m} = \frac{-e}{2}$$

$$\Rightarrow \lim_{\alpha \rightarrow 0} \frac{e^{\left[\cos \alpha^n - 1 \right]}}{\cos \alpha^n - 1} \times \frac{\cos \alpha^n - 1}{\alpha^m} = \frac{-e}{2}$$

$$\Rightarrow e \lim_{\alpha \rightarrow 0} \frac{-2 \sin^2 \frac{\alpha^n}{2} \times \left(\frac{\alpha^n}{2} \right)^2}{\left(\frac{\alpha^n}{2} \right)^2} \times \frac{1}{\alpha^m} = \frac{-e}{2}$$

$$\Rightarrow \frac{-e}{2} \alpha^{2n-m} = \frac{-e}{2} \text{ or } \alpha^{2n-m} = 1$$

$$\Rightarrow 2n - m = 0 \Rightarrow \frac{m}{n} = 2$$

$$5. (7) \lim_{x \rightarrow 0} \frac{x^2 \sin \beta x}{\alpha x - \sin x} = 1$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{x^3 \beta}{\alpha x - \sin x} = 1$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{x^3 \beta}{\alpha x - \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \infty \right)} = 1$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{x^3 \beta}{(\alpha - 1)x + \frac{x^3}{3!} - \frac{x^5}{5!} + \dots \infty} = 1$$

For above to be possible, we should have

$$\alpha - 1 = 0 \text{ and } \beta = \frac{1}{3!}$$

$$\Rightarrow \alpha = 1 \text{ and } \beta = \frac{1}{6}$$

$$\therefore 6(\alpha + \beta) = 6 \left(1 + \frac{1}{6} \right) = 7$$

Section-B

JEE Main/ AIEEE

$$1. \quad (d) \quad \lim_{x \rightarrow 0} \frac{\sqrt{1 - \cos 2x}}{\sqrt{2x}} \Rightarrow \lim_{x \rightarrow 0} \frac{\sqrt{1 - (1 - 2 \sin^2 x)}}{\sqrt{2x}};$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{2 \sin^2 x}}{\sqrt{2x}} \Rightarrow \lim_{x \rightarrow 0} \frac{|\sin x|}{x}$$

The limit of above does not exist as
LHS = -1 $\neq$ RHL = 1

$$2. \quad (a) \quad \lim_{x \rightarrow \infty} \left(\frac{x^2 + 5x + 3}{x^2 + x + 2} \right)^x = \lim_{x \rightarrow \infty} \left(1 + \frac{4x + 1}{x^2 + x + 2} \right)^x$$

$$= \lim_{x \rightarrow \infty} \left[\left(1 + \frac{4x + 1}{x^2 + x + 2} \right)^{\frac{x^2 + x + 2}{4x + 1}} \right]^{\frac{(4x + 1)x}{x^2 + x + 2}}$$

$$= e^{\lim_{x \rightarrow \infty} \frac{4x^2 + x}{x^2 + x + 2} \left[\lim_{x \rightarrow \infty} \left(1 + \frac{1}{\lambda x} \right)^{\frac{1}{\lambda}} = e^{\lambda} \right]}$$

$$= e^{\lim_{x \rightarrow \infty} \frac{4 + \frac{1}{x}}{1 + \frac{1}{x} + \frac{2}{x^2}}} = e^4$$

3. (c) Apply L H Rule

$$\text{We have, } \lim_{x \rightarrow 2} \frac{xf(2) - 2f(x)}{x - 2} \quad \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 2} \frac{f(2) - 2f'(x)}{1} = \frac{f(2) - 2f'(2)}{1}$$

$$= 4 - 2 \times 4 = -4.$$

$$4. \quad (a) \quad \text{We have } \lim_{n \rightarrow \infty} \frac{1^p + 2^p + \dots + n^p}{n^{p+1}};$$

$$\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{r^p}{n^p \cdot n} = \int_0^1 x^p dx = \left[\frac{x^{p+1}}{p+1} \right]_0^1 = \frac{1}{p+1}$$

5. (d) Since $\lim_{x \rightarrow 0} [x]$ does not exist, hence the required limit does not exist.

$$6. \quad (a) \quad \lim_{x \rightarrow 1} \frac{\sqrt{f(x)} - 1}{\sqrt{x} - 1} \quad \left(\frac{0}{0} \right) \text{ form using L' Hospital's rule}$$

$$= \lim_{x \rightarrow 1} \frac{\frac{1}{2\sqrt{f(x)}} f'(x)}{\frac{1}{2\sqrt{x}}} = \frac{f'(1)}{\sqrt{f(1)}} = \frac{2}{1} = 2.$$

7. (b) Let a is a rational number other than 0, in $[-5, 5]$, then

$$f(a) = a \text{ and } \lim_{x \rightarrow a} f(x) = -a$$

[As in the immediate neighbourhood of a rational number, we find irrational numbers]

$\therefore f(x)$ is not continuous at any rational number

If a is irrational number, then

$$f(a) = -a \text{ and } \lim_{x \rightarrow a} f(x) = a$$

$\therefore f(x)$ is not continuous at any irrational number clearly

$$\lim_{x \rightarrow 0} f(x) = f(0) = 0$$

$\therefore f(x)$ is continuous at $x = 0$

$$8. \quad (d) \quad \therefore f''(x) - g''(x) = 0$$

Integrating, $f'(x) - g'(x) = c$;

$$\Rightarrow f'(1) - g'(1) = c \Rightarrow 4 - 2 = c \Rightarrow c = 2.$$

$$\therefore f'(x) - g'(x) = 2;$$

Integrating, $f(x) - g(x) = 2x + c_1$

$$\Rightarrow f(2) - g(2) = 4 + c_1 \Rightarrow 9 - 3 = 4 + c_1;$$

$$\Rightarrow c_1 = 2 \quad \therefore f(x) - g(x) = 2x + 2$$

$$\text{At } x = 3/2, f(x) - g(x) = 3 + 2 = 5.$$

$$9. \quad (c) \quad f(x+y) = f(x) \times f(y)$$

Differentiate with respect to x , treating y as constant

$$f'(x+y) = f'(x)f(y)$$

Putting $x = 0$ and $y = x$, we get $f''(x) = f'(0)f(x)$;

$$\Rightarrow f'(5) = 3f(5) = 3 \times 2 = 6.$$

10. (a) The given expression can be written as

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n} \right)^4 - \lim_{n \rightarrow \infty} \frac{1}{n} \cdot \lim_{n \rightarrow \infty} \frac{1}{n} \left(\frac{r}{n} \right)^3$$

$$= \int_0^1 x^4 dx - \lim_{n \rightarrow \infty} \frac{1}{n} \times \int_0^1 x^3 dx = \left[\frac{x^5}{5} \right]_0^1 - 0 = \frac{1}{5}$$

$$11. \quad (d) \quad \lim_{x \rightarrow 0} \frac{\log(3+x) - \log(3-x)}{x} = K \quad (\text{by L'Hospital rule})$$

$$\lim_{x \rightarrow 0} \frac{\frac{1}{3+x} - \frac{-1}{3-x}}{1} = K \quad \therefore \frac{2}{3} = K$$

$$12. \quad (d) \quad \lim_{x \rightarrow 0} \frac{\frac{d}{dx} \int_0^{x^2} \sec^2 t dt}{\frac{d}{dx} (x \sin x)} = \lim_{x \rightarrow 0} \frac{\sec^2 x^2 \cdot 2x}{\sin x + x \cos x}$$

(by L' Hospital rule)

$$\lim_{x \rightarrow 0} \frac{2 \sec^2 x^2}{\left(\frac{\sin x}{x} + \cos x \right)} = \frac{2 \times 1}{1 + 1} = 1$$

$$13. \quad (b) \quad \lim_{x \rightarrow a} \frac{f(a)g'(x) - g(a)f'(x)}{g'(x) - f'(x)} = 4$$

(By L' Hospital rule)

$$\lim_{x \rightarrow a} \frac{k g'(x) - k f'(x)}{g'(x) - f'(x)} = 4 \quad \therefore k = 4.$$

$$14. (d) \lim_{x \rightarrow \frac{\pi}{2}} \frac{\tan\left(\frac{\pi}{4} - \frac{x}{2}\right) \cdot (1 - \sin x)}{(\pi - 2x)^3}$$

$$\text{Let } x = \frac{\pi}{2} + y; y \rightarrow 0$$

$$= \lim_{y \rightarrow 0} \frac{\tan\left(-\frac{y}{2}\right) \cdot (1 - \cos y)}{(-2y)^3} = \lim_{y \rightarrow 0} \frac{-\tan \frac{y}{2} \cdot 2 \sin^2 \frac{y}{2}}{(-8) \cdot \frac{y^3}{8} \cdot 8}$$

$$= \lim_{y \rightarrow 0} \frac{1}{32} \frac{\tan \frac{y}{2}}{\left(\frac{y}{2}\right)} \cdot \left[\frac{\sin y/2}{y/2}\right]^2 = \frac{1}{32}$$

$$15. (c) f(0) = 0; f(x) = xe^{-\left(\frac{1}{|x|} + \frac{1}{x}\right)}$$

$$\text{R.H.L. } \lim_{h \rightarrow 0} (0+h)e^{-2/h} = \lim_{h \rightarrow 0} \frac{h}{e^{2/h}} = 0$$

$$\text{L.H.L. } \lim_{h \rightarrow 0} (0-h)e^{-\left(\frac{1}{h} - \frac{1}{h}\right)} = 0$$

therefore, $f(x)$ is continuous.

$$\text{R.H.D.} = \lim_{h \rightarrow 0} \frac{(0+h)e^{-\left(\frac{1}{h} + \frac{1}{h}\right)} - 0}{h} = 0$$

$$\text{L.H.D.} = \lim_{h \rightarrow 0} \frac{(0-h)e^{-\left(\frac{1}{h} - \frac{1}{h}\right)} - 0}{-h} = 1$$

therefore, L.H.D. $\neq$ R.H.D.

$f(x)$ is not differentiable at $x = 0$.

$$16. (b) \text{ We know that } \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$$

$$\therefore \lim_{x \rightarrow \infty} \left(1 + \frac{a}{x} + \frac{b}{x^2}\right)^{2x} = e^2$$

$$\Rightarrow \lim_{x \rightarrow \infty} \left[\left(1 + \frac{a}{x} + \frac{b}{x^2}\right)^{\left(\frac{1}{\frac{a}{x} + \frac{b}{x^2}}\right)} \right]^{2x \left(\frac{a}{x} + \frac{b}{x^2}\right)} = e^2$$

$$\Rightarrow e^{\lim_{x \rightarrow \infty} 2 \left[\frac{a+b}{x} \right]} = e^2 \Rightarrow e^{2a} = e^2 \Rightarrow a = 1 \text{ and } b \in R$$

$$17. (c) f(x) = \frac{1 - \tan x}{4x - \pi} \text{ is continuous in } \left[0, \frac{\pi}{2}\right]$$

$$\therefore f\left(\frac{\pi}{4}\right) = \lim_{x \rightarrow \frac{\pi}{4}} f(x) = \lim_{x \rightarrow \frac{\pi}{4}} f(x) = \lim_{h \rightarrow 0} f\left(\frac{\pi}{4} + h\right)$$

$$= \lim_{h \rightarrow 0} \frac{1 - \tan\left(\frac{\pi}{4} + h\right)}{4\left(\frac{\pi}{4} + h\right) - \pi}, h > 0 = \lim_{h \rightarrow 0} \frac{1 - \frac{1 + \tan h}{1 - \tan h}}{4h}$$

$$= \lim_{h \rightarrow 0} \frac{-2}{1 - \tanh} \cdot \frac{\tanh h}{4h} = \frac{-2}{4} = -\frac{1}{2}$$

$$18. (d) \lim_{n \rightarrow \infty} \left[\frac{1}{n^2} \sec^2 \frac{1}{n^2} + \frac{2}{n^2} \sec^2 \frac{4}{n^2} + \frac{3}{n^2} \sec^2 \frac{9}{n^2} + \dots + \frac{1}{n} \sec^2 1 \right]$$

$$= \lim_{n \rightarrow \infty} \frac{r}{n^2} \sec^2 \frac{r^2}{n^2} = \lim_{n \rightarrow \infty} \frac{1}{n} \cdot \frac{r}{n} \sec^2 \frac{r^2}{n^2}$$

$\Rightarrow$ Given limit is equal to value of integral

$$\int_0^1 x \sec^2 x^2 dx$$

$$\text{or } \frac{1}{2} \int_0^1 2x \sec^2 x^2 dx = \frac{1}{2} \int_0^1 \sec^2 t dt \quad [\text{put } x^2 = t]$$

$$= \frac{1}{2} (\tan t)_0^1 = \frac{1}{2} \tan 1.$$

$$19. (a) \text{ Given limit} = \lim_{x \rightarrow \alpha} \frac{1 - \cos a(x - \alpha)(x - \beta)}{(x - \alpha)^2}$$

$$= \lim_{x \rightarrow \alpha} \frac{2 \sin^2 \left(a \frac{(x - \alpha)(x - \beta)}{2} \right)}{(x - \alpha)^2}$$

$$= \lim_{x \rightarrow \alpha} \frac{2}{(x - \alpha)^2} \times \frac{\sin^2 \left(a \frac{(x - \alpha)(x - \beta)}{2} \right)}{\frac{a^2 (x - \alpha)^2 (x - \beta)^2}{4}} \times \frac{a^2 (x - \alpha)^2 (x - \beta)^2}{4}$$

$$= \frac{a^2 (\alpha - \beta)^2}{2}.$$

$$20. (c) f'(1) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h};$$

As function is differentiable so it is continuous as it

is given that $\lim_{h \rightarrow 0} \frac{f(1+h)}{h} = 5$ and hence $f(1) = 0$

$$\text{Hence } f'(1) = \lim_{h \rightarrow 0} \frac{f(1+h)}{h} = 5$$

$$21. (a) \text{ As } f(1) = -2 \text{ \& } f'(x) \geq 2 \forall x \in [1, 6]$$

Applying Lagrange's mean value theorem

$$\frac{f(6) - f(1)}{5} = f'(c) \geq 2 \Rightarrow f(6) \geq 10 + f(1)$$

$$\Rightarrow f(6) \geq 10 - 2 \Rightarrow f(6) \geq 8.$$

$$22. (b) f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$|f'(x)| = \lim_{h \rightarrow 0} \left| \frac{f(x+h) - f(x)}{h} \right| \leq \lim_{h \rightarrow 0} \left| \frac{(h)^2}{h} \right|$$

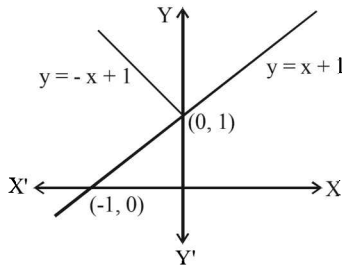
$$\Rightarrow |f'(x)| \leq 0 \Rightarrow f'(x) = 0$$

$$\Rightarrow f(x) = \text{constant}$$

$$\text{As } f(0) = 0 \Rightarrow f(1) = 0$$

Limits, Continuity and Differentiability

23. (a) $f(x) = \min \{x+1, |x|+1\} \Rightarrow f(x) = x+1 \nrightarrow x \in R$



Hence, $f(x)$ is differentiable everywhere for all $x \in R$.

24. (b) Given, $f(x) = \frac{1}{x} - \frac{2}{e^{2x}-1} \Rightarrow f(0) = \lim_{x \rightarrow 0} \frac{1}{x} - \frac{2}{e^{2x}-1}$

$$= \lim_{x \rightarrow 0} \frac{(e^{2x}-1)-2x}{x(e^{2x}-1)} \quad \left[\frac{0}{0} \text{ form} \right]$$

$\therefore$ using, L'Hospital rule

$$f(0) = \lim_{x \rightarrow 0} \frac{4e^{2x}}{2(xe^{2x} + e^{2x} \cdot 1) + e^{2x} \cdot 2}$$

$$= \lim_{x \rightarrow 0} \frac{4e^{2x}}{4xe^{2x} + 2e^{2x} + 2e^{2x}} \quad \left[\frac{0}{0} \text{ form} \right]$$

$$= \lim_{x \rightarrow 0} \frac{4e^{2x}}{4(xe^{2x} + e^{2x})} = \frac{4 \cdot e^0}{4(0 + e^0)} = 1$$

25. (c) We have $f(x) = \begin{cases} (x-1)\sin\left(\frac{1}{x-1}\right), & \text{if } x \neq 1 \\ 0, & \text{if } x = 1 \end{cases}$

$$Rf'(1) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h \sin \frac{1}{h} - 0}{h} = \lim_{h \rightarrow 0} \sin \frac{1}{h} = \text{a finite number}$$

Let this finite number be l

$$L f'(1) = \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{-h \sin\left(\frac{1}{-h}\right)}{-h} = \lim_{h \rightarrow 0} \sin\left(\frac{1}{-h}\right)$$

$$= -\lim_{h \rightarrow 0} \sin\left(\frac{1}{h}\right) = -(\text{a finite number}) = -l$$

Thus $Rf'(1) \neq Lf'(1)$

$\therefore f$ is not differentiable at $x = 1$

$$\text{Also, } f'(0) = \sin \frac{1}{(x-1)} - \frac{x-1}{(x-1)^2} \cos\left(\frac{1}{x-1}\right) \Bigg|_{x=0}$$

$$= -\sin 1 + \cos 1 \therefore f \text{ is differentiable at } x = 0$$

26. (d) $f(x)$ is a positive increasing function.

$$\therefore 0 < f(x) < f(2x) < f(3x)$$

$$\Rightarrow 0 < 1 < \frac{f(2x)}{f(x)} < \frac{f(3x)}{f(x)}$$

$$\Rightarrow \lim_{x \rightarrow \infty} 1 \leq \lim_{x \rightarrow \infty} \frac{f(2x)}{f(x)} \leq \lim_{x \rightarrow \infty} \frac{f(3x)}{f(x)}$$

By Sandwich Theorem.

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{f(2x)}{f(x)} = 1$$

27. (d) $\lim_{x \rightarrow 2} \frac{\sqrt{1 - \cos\{2(x-2)\}}}{x-2} = \lim_{x \rightarrow 2} \frac{\sqrt{2} |\sin(x-2)|}{x-2}$

$$\text{L.H.L.} = \lim_{x \rightarrow 2^-} \frac{\sqrt{2} \sin(x-2)}{(x-2)} = -\sqrt{2}$$

$$\text{R.H.L.} = \lim_{x \rightarrow 2^+} \frac{\sqrt{2} \sin(x-2)}{(x-2)} = \sqrt{2}$$

Thus L.H.L. $\neq$ R.H.L.

Hence, $\lim_{x \rightarrow 2} \frac{\sqrt{1 - \cos\{2(x-2)\}}}{x-2}$ does not exist.

28. (b) $L.H.L = \lim_{x \rightarrow 0^-} f(x) = \lim_{h \rightarrow 0} \frac{\sin\{(p+1)(-h)\} - \sin(-h)}{-h}$

$$= \lim_{h \rightarrow 0} \frac{-\sin(p+1)h}{-h} + \frac{\sin(-h)}{-h} = p+1+1 = p+2$$

$$R.H.L. = \lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0} \frac{\sqrt{1+h}-1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{1}{(\sqrt{1+h}+1)} = \frac{1}{2}$$

$$\text{and } f(0) = q \Rightarrow p = -\frac{3}{2}, q = \frac{1}{2}$$

29. (d) $\lim_{x \rightarrow 5} \frac{(f(x))^2 - 9}{\sqrt{|x-5|}} = 0$

$$\lim_{x \rightarrow 5} [(f(x))^2 - 9] = 0 \Rightarrow \lim_{x \rightarrow 5} f(x) = 3$$

30. (a) Let $f(x) = [x] \cos\left(\frac{2x-1}{2}\right)$

Doubtful points are $x = n, n \in I$

$$\text{L.H.L.} = \lim_{x \rightarrow n^-} [x] \cos\left(\frac{2x-1}{2}\right) \pi$$

$$= (n-1) \cos\left(\frac{2n-1}{2}\right) \pi = 0$$

($\because [x]$ is the greatest integer function)

$$\text{R.H.L.} = \lim_{x \rightarrow n^+} [x] \cos\left(\frac{2x-1}{2}\right) \pi = n \cos\left(\frac{2n-1}{2}\right) \pi = 0$$

Now, value of the function at $x = n$ is $f(n) = 0$

Since, L.H.L. = R.H.L. = $f(n)$

$\therefore f(x) = [x] \cos\left(\frac{2x-1}{2}\right)$ is continuous for every real x .

31. (c) $f(x) = |x-2| = \begin{cases} x-2, & x-2 \geq 0 \\ 2-x, & x-2 \leq 0 \end{cases}$

$$= \begin{cases} x-2, & x \geq 2 \\ 2-x, & x \leq 2 \end{cases}$$

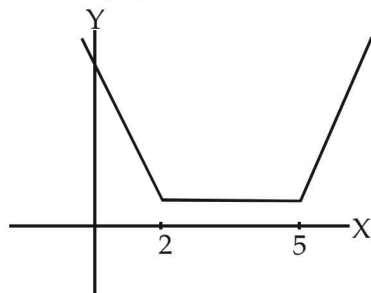
Similarly, $f(x) = |x-5| = \begin{cases} x-5, & x \geq 5 \\ 5-x, & x \leq 5 \end{cases}$

$$\therefore f(x) = |x-2| + |x-5| = \begin{cases} x-2+5-x=3, & 2 \leq x \leq 5 \end{cases}$$

Thus $f(x) = 3, 2 \leq x \leq 5$

$$f'(x) = 0, 2 < x < 5$$

$$f'(4) = 0 \therefore \text{statement 1 is true.}$$



$$\therefore f(2) = 0 + |2-5| = 3 \text{ and } f(5) = |5-2| + 0 = 3$$

$\therefore$ statement-2 is also true and a correct explanation for statement 1.

32. (d) Multiply and divide by x in the given expression, we get

$$\lim_{x \rightarrow 0} \frac{(1 - \cos 2x)}{x^2} \cdot \frac{(3 + \cos x)}{1} \cdot \frac{x}{\tan 4x}$$

$$= \lim_{x \rightarrow 0} \frac{2 \sin^2 x}{x^2} \cdot \frac{3 + \cos x}{1} \cdot \frac{x}{\tan 4x}$$

$$= 2 \lim_{x \rightarrow 0} \frac{\sin^2 x}{x^2} \cdot \lim_{x \rightarrow 0} (3 + \cos x) \cdot \lim_{x \rightarrow 0} \frac{x}{\tan 4x}$$

$$= 2.4 \cdot \frac{1}{4} \lim_{x \rightarrow 0} \frac{4x}{\tan 4x} = 2.4 \cdot \frac{1}{4} = 2$$

33. (b) Consider $\lim_{x \rightarrow 0} \frac{\sin(\pi \cos^2 x)}{x^2}$

$$= \lim_{x \rightarrow 0} \frac{\sin[\pi(1 - \sin^2 x)]}{x^2}$$

$$= \lim_{x \rightarrow 0} \sin \frac{(\pi - \pi \sin^2 x)}{x^2} \quad [\because \sin(\pi - \theta) = \sin \theta]$$

$$= \lim_{x \rightarrow 0} \sin \frac{(\pi \sin^2 x)}{\pi \sin^2 x} \times \frac{\pi \sin^2 x}{x^2}$$

$$= \lim_{x \rightarrow 0} 1 \times \pi \left(\frac{\sin x}{x} \right)^2 = \pi$$

34. (a) Multiply and divide by x in the given expression, we get

$$\lim_{x \rightarrow 0} \frac{(1 - \cos 2x)}{x^2} \cdot \frac{(3 + \cos x)}{1} \cdot \frac{x}{\tan 4x}$$

$$= \lim_{x \rightarrow 0} \frac{2 \sin^2 x}{x^2} \cdot \frac{3 + \cos x}{1} \cdot \frac{x}{\tan 4x}$$

$$= 2 \lim_{x \rightarrow 0} \frac{\sin^2 x}{x^2} \cdot \lim_{x \rightarrow 0} (3 + \cos x) \cdot \lim_{x \rightarrow 0} \frac{x}{\tan 4x}$$

$$= 2.4 \cdot \frac{1}{4} \lim_{x \rightarrow 0} \frac{4x}{\tan 4x} = 2.4 \cdot \frac{1}{4} = 2$$

35. (c) Since $g(x)$ is differentiable, it will be continuous at $x = 3$

$$\therefore \lim_{x \rightarrow 3^-} g(x) = \lim_{x \rightarrow 3^+} g(x)$$

$$2k = 3m + 2 \quad \dots(1)$$

Also $g(x)$ is differentiable at $x = 0$

$$\therefore \lim_{x \rightarrow 3^-} g'(x) = \lim_{x \rightarrow 3^+} g'(x)$$

$$\frac{K}{2\sqrt{3+1}} = m$$

$$k = 4m \quad \dots(2)$$

Solving (1) and (2), we get

$$m = \frac{2}{5}, \quad k = \frac{8}{5}$$

$$\therefore k + m = 2$$

36. (d) $g(x) = f(f(x))$

In the neighbourhood of $x = 0$,

$$f(x) = |\log 2 - \sin x| = (\log 2 - \sin x)$$

$$\therefore g(x) = |\log 2 - \sin| \log 2 - \sin x ||$$

$$= (\log 2 - \sin(\log 2 - \sin x))$$

$\therefore g(x)$ is differentiable

$$\text{and } g'(x) = -\cos(\log 2 - \sin x) (-\cos x)$$

$$\Rightarrow g'(0) = \cos(\log 2)$$

$$37. \quad (d) \quad y = \lim_{n \rightarrow \infty} \left(\frac{(n+1)(n+2)\dots 3n}{n^{2n}} \right)^{\frac{1}{n}}$$

$$\ln y = \lim_{n \rightarrow \infty} \frac{1}{n} \ln \left(1 + \frac{1}{n} \right) \left(1 + \frac{2}{n} \right) \dots \left(1 + \frac{2n}{n} \right)$$

$$\ln y = \lim_{n \rightarrow \infty} \frac{1}{n} \left[\ln \left(1 + \frac{1}{n} \right) + \ln \left(1 + \frac{2}{n} \right) + \dots + \ln \left(1 + \frac{2n}{n} \right) \right]$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{2n} \ln \left(1 + \frac{r}{n} \right)$$

$$= \int_0^2 \ln(1+x) dx$$

$$\text{Let } 1+x=t \Rightarrow dx=dt$$

$$\text{when } x=0, \quad t=1$$

$$x=2, \quad t=3$$

$$\ln y = \int_1^3 \ln t \, dt = [t \ln t - t]_1^3 = \ln \left(\frac{3^3}{e^2} \right) = \ln \left(\frac{27}{e^2} \right)$$

$$\Rightarrow y = \frac{27}{e^2}$$

$$38. \quad (a) \quad \ln P = \lim_{x \rightarrow 0^+} \frac{1}{2x} \ln(1 + \tan^2 \sqrt{x})$$

$$\lim_{x \rightarrow 0^+} \frac{1}{x} \ln(\sec \sqrt{x})$$

Applying L hospital's rule :

$$= \lim_{x \rightarrow 0^+} \frac{\sec \sqrt{x} \tan \sqrt{x}}{\sec \sqrt{x} \cdot 2\sqrt{x}}$$

$$= \lim_{x \rightarrow 0^+} \frac{\tan \sqrt{x}}{2\sqrt{x}}$$

$$= \frac{1}{2}$$